



Robotic Grasping and Manipulation

**- robotic hand design
(from rigid to soft)**

Miao Li

2025, Wuhan University

WeChat: 15527576906

Email: limiao712@gmail.com

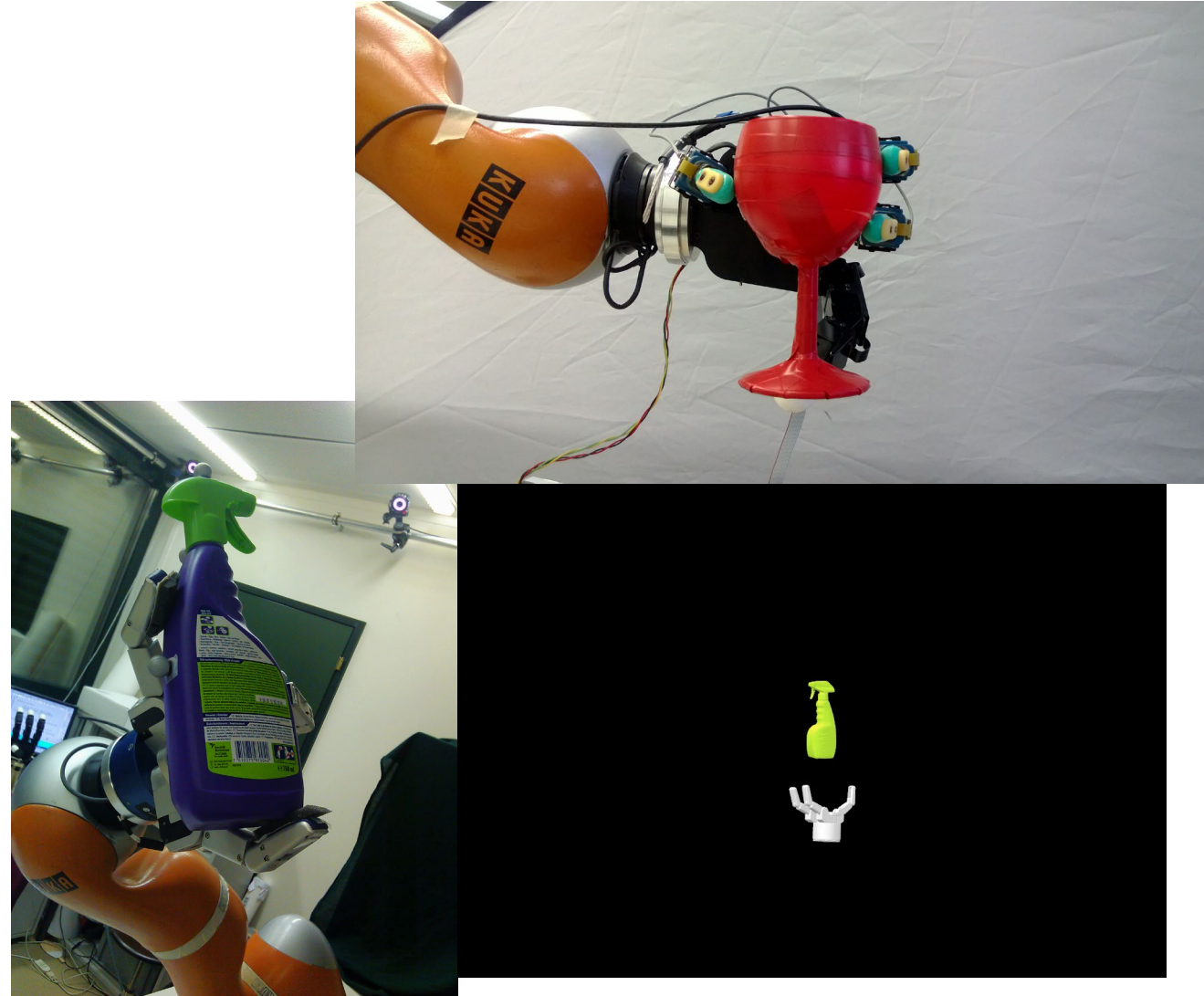


- Please select your project ASAP
- Try to use **5-10 slides** to explain your motivation, your brief concept, and why is interesting, etc ...
- Think outside the box!
- Simple, robust, easy to fabricate



What is important for robotic grasping?

- Hand design
- Grasp planning
- Motion Planning
- Grasp control
- Learning
- Tactile sensing





Key components of a **new** hand

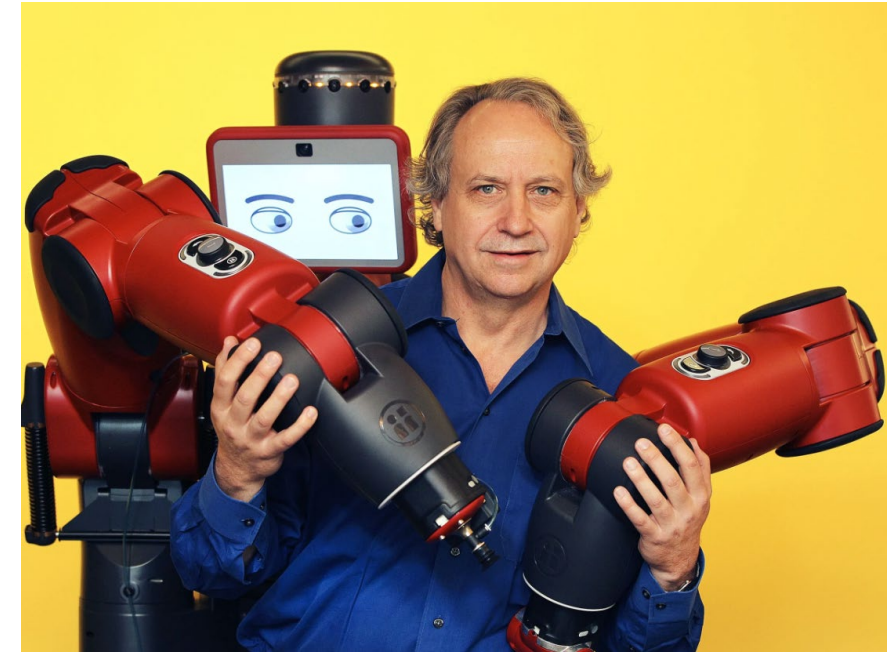
- **Body material**
- **Motion mechanism**
- **Actuator**
- **Sensor integrated**
- **Control algorithm**
- **Application scenarios**



Can be “new” in any sense



Hand-centric approach



RODNEY BROOKS *Robots, AI, and other stuff* **BLOG** MIT ROBUST.AI



POST: WHY TODAY'S HUMANOIDS WON'T LEARN DEXTERITY

SEPTEMBER 26, 2025 — ESSAYS

Why Today's Humanoids Won't Learn Dexterity 🦋

rodneybrooks.com/why-todays-humanoids-wont-learn-dexterity/

Search ...

<https://rodneybrooks.com/why-todays-humanoids-wont-learn-dexterity/>

Dexterous vs. Gripper



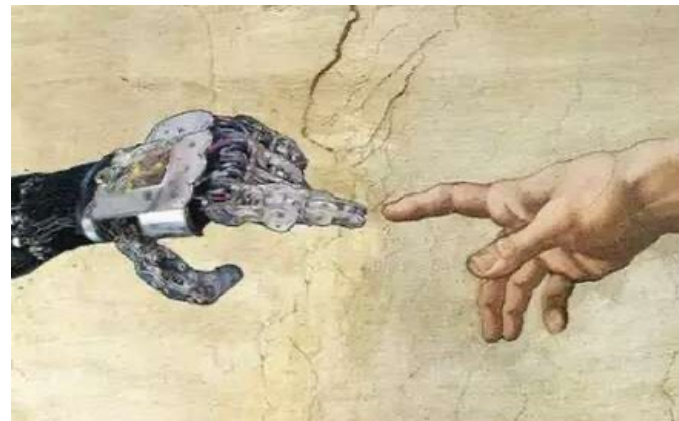


Hands for Dexterous Manipulation and Robust Grasping: A Difficult Road Toward Simplicity

Antonio Bicchi, *Senior Member, IEEE*

I. INTRODUCTION

Ἀναξαγόρας μὲν οὖν φησι διὰ τὸ χεῖρας ἔχειν φρονιμώτατον εἶναι τῶν ζώων ἀνθρώπων: εὐλογον δὲ διὰ τὸ φρονιμώτατον εἶναι χεῖρας λαμβάνειν.”¹ In one of his books on nature sciences [6], the greek philosopher Aristotle (384–322 BC) thus argued against the conceptions of his late colleague Anaxagoras (500?–428 BC) regarding the relationship between human hands and mind. As they appear to be the two most distinguished features of humans among animals, the two philosophers debated whether it was because humans had dexterous hands that they became intelligent, or the other way around. Anaxagoras’ intuition has been later on confirmed by several findings of paleoanthropologists, showing that the mechanical dexterity of the human hand has been a major factor in allowing homo sapiens to develop a superior brain (a similar role played by the anatomical structure of the human larynx in relation with speech capabilities has been also recognized).



Antonio Bicchi (S’87–M’89–SM’99) was born in Toscana, Italy, in 1959. He received the Laurea degree from the University of Pisa, Pisa, Italy, in 1984, and graduated from the University of Bologna, Bologna, Italy, in 1988.

From 1988 to 1990, he was a Postdoctoral Scholar with the Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge. He is currently an Associate Professor of Systems Theory and Robotics with the Department of Electrical Systems and Automation (DSEA), University of Pisa. Since 1990, he has been leading the Robotics Group at the Interdepartmental Research Center “E.

Piaggio,” University of Pisa. His main research interests within robotics are in dextrous manipulation, as follows: force–torque and tactile sensing, haptics, and sensory control; dynamics, kinematics, and control of complex mechanical systems; and motion planning and control for nonholonomic and quantized systems.



Hand-centric approach

**AR ANNUAL
REVIEWS**

*Annual Review of Control, Robotics, and
Autonomous Systems*

A Century of Robotic Hands

C. Piazza,¹ G. Grioli,² M.G. Catalano,² and A. Bicchi^{1,2}

¹Centro di Ricerca "E. Piaggio" and Dipartimento di Ingegneria Informatica, Università di Pisa,
56122 Pisa, Italy; email: cristina.piazza@ing.unipi.it

²Soft Robotics for Human Cooperation and Rehabilitation, Istituto Italiano di Tecnologia,
16163 Genova, Italy

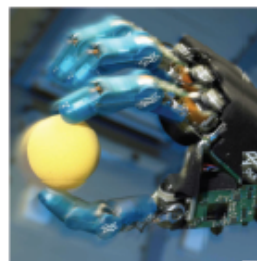


<https://www.annualreviews.org/docserver/fulltext/control/2/1/annurev-control-060117-105003.pdf?expires=1774437735&id=id&accname=guest&checksum=95BF10E1D031220C5FBB5C7BB8441568>

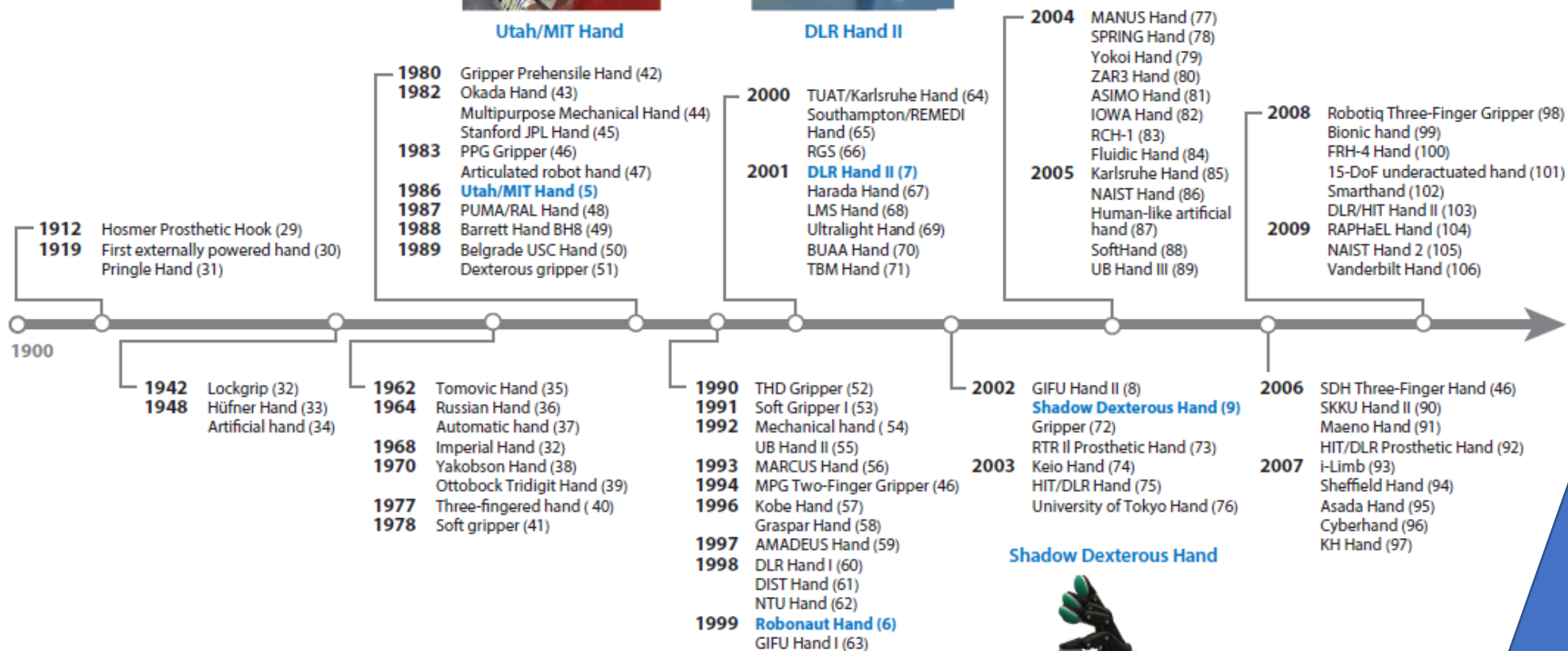
<https://soundcloud.com/ieeeras-softrobotics/soft-robotics-with-antonio-bicchi>



Utah/MIT Hand



DLR Hand II

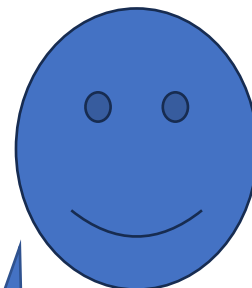


Robonaut Hand



Shadow Dexterous Hand

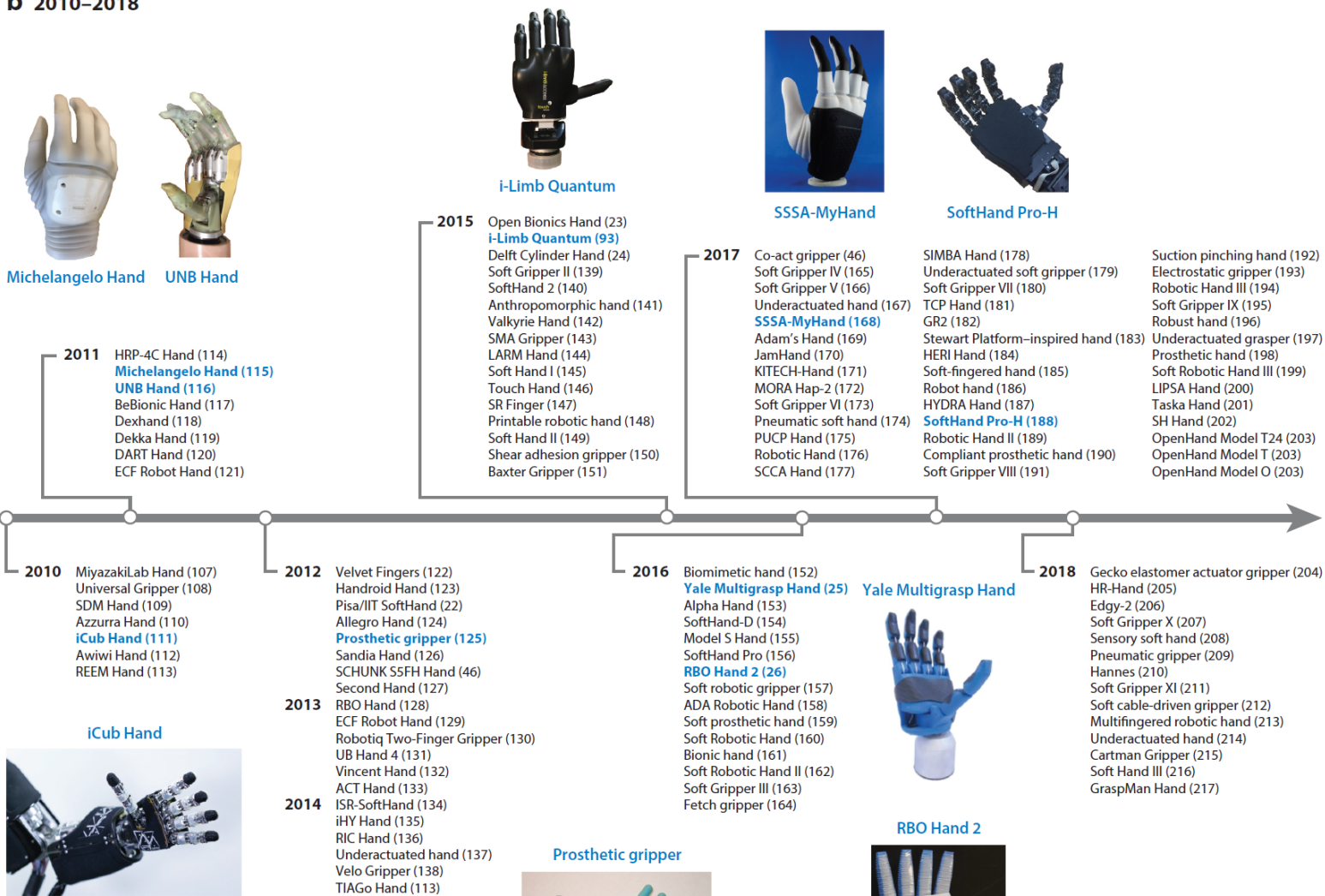
What ?





Hand-centric approach

b 2010–2018

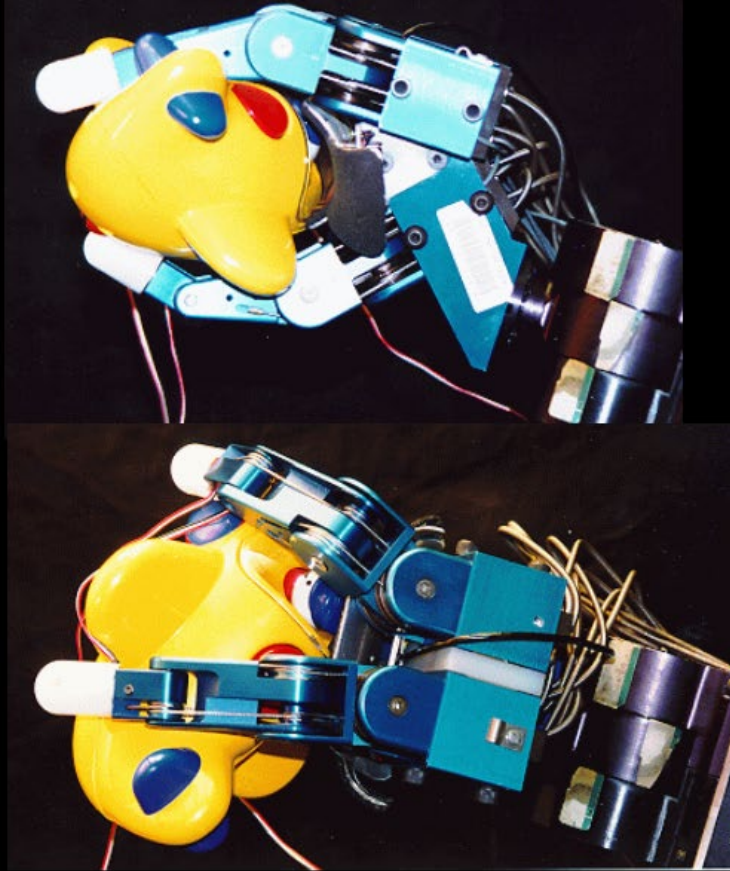
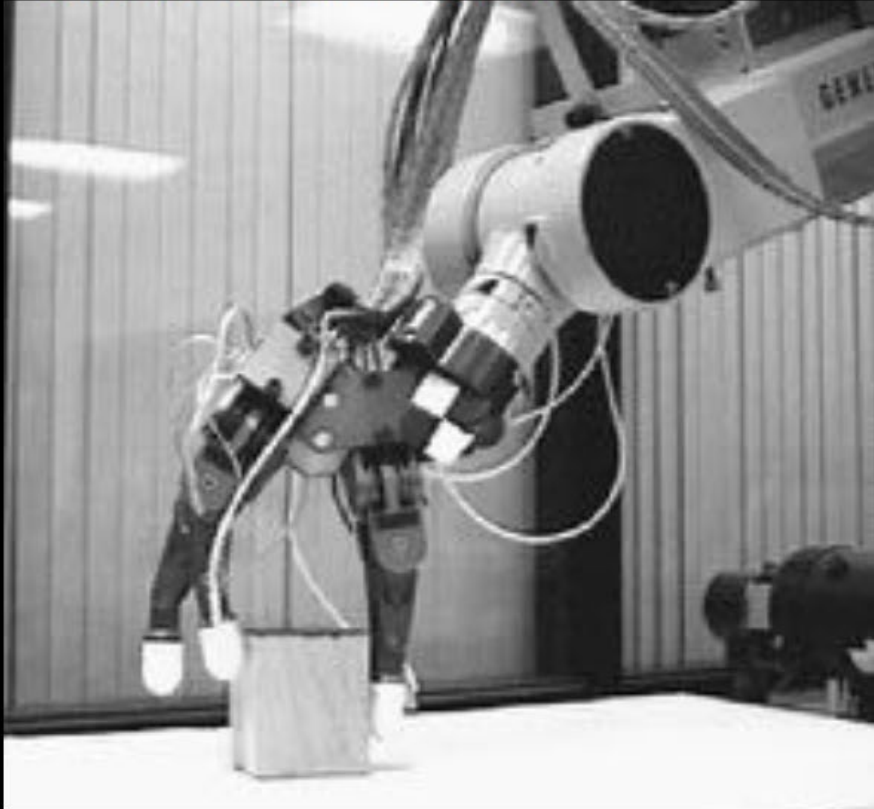


What ?

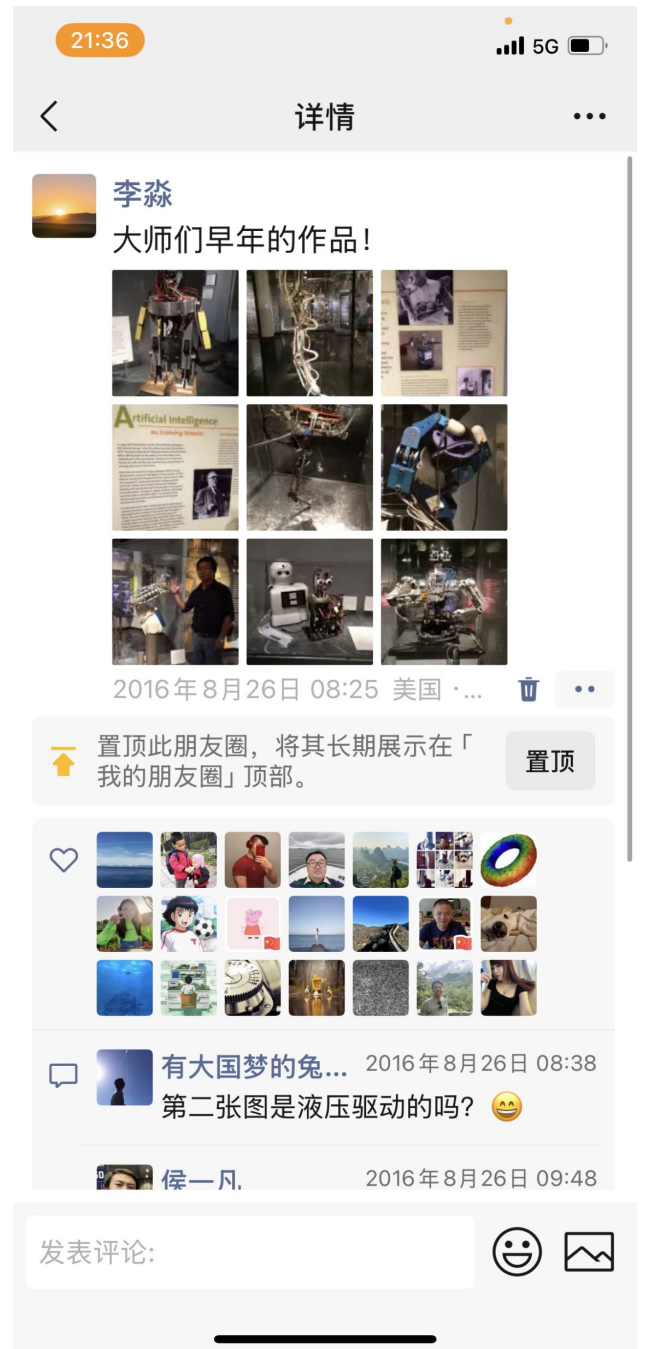


Robotic Hands

Stanford / JPL hand

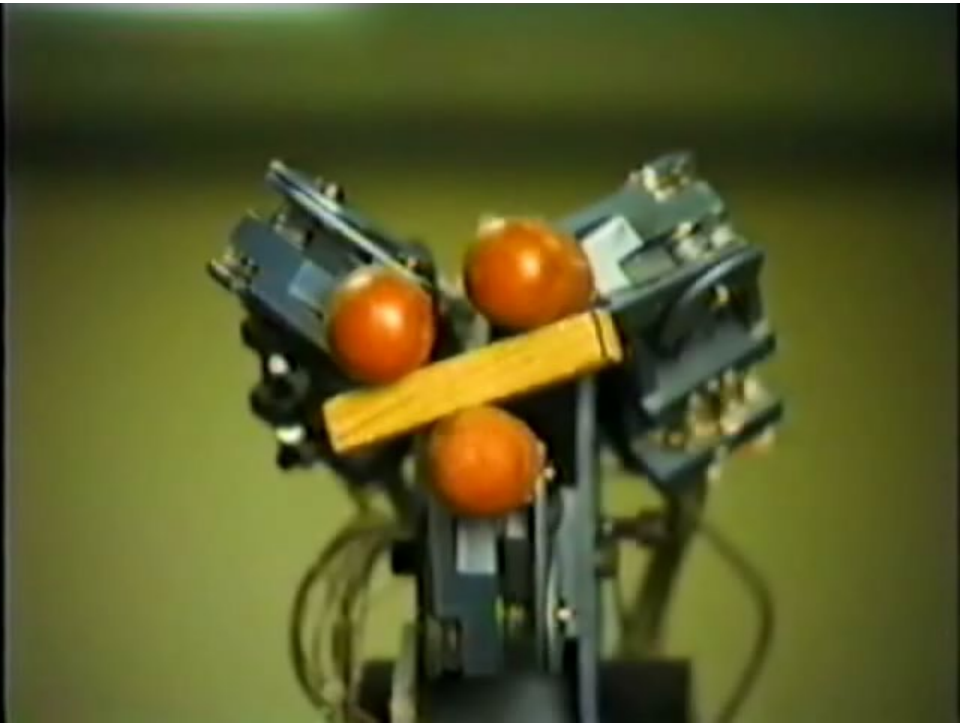


9 DoF, 4 tendons/finger, designed for fingertip manipulation
Strain gauge fingertip sensors

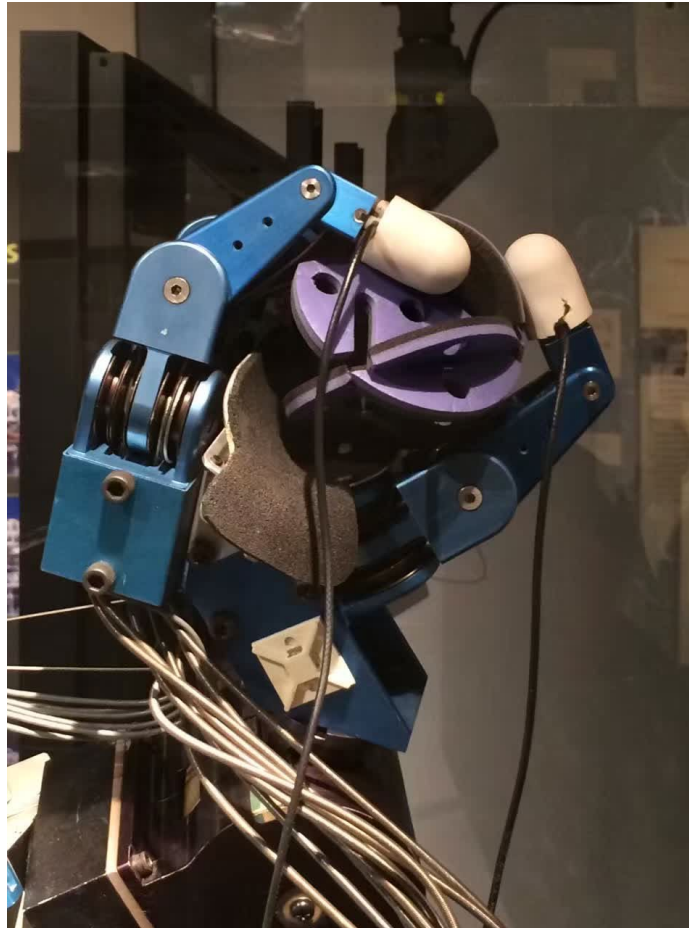




Robotic Hands



40+ years ago





Robotic Hands

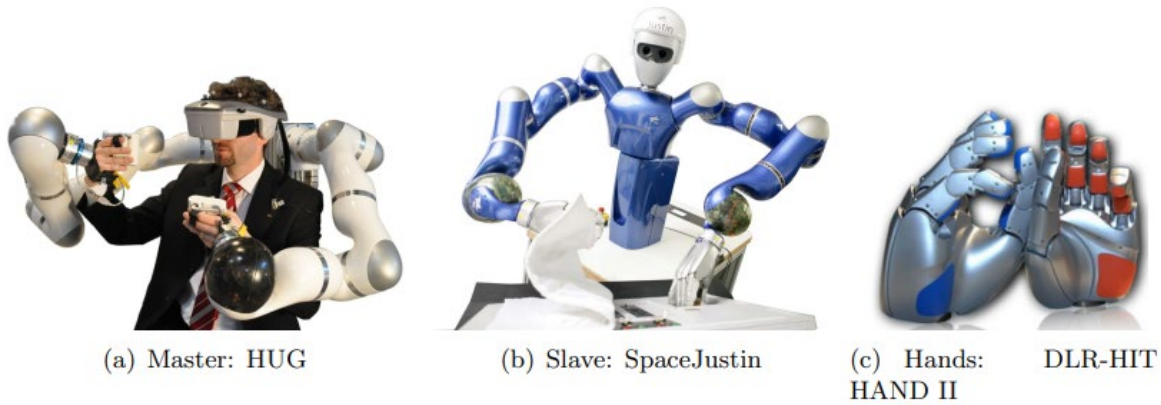
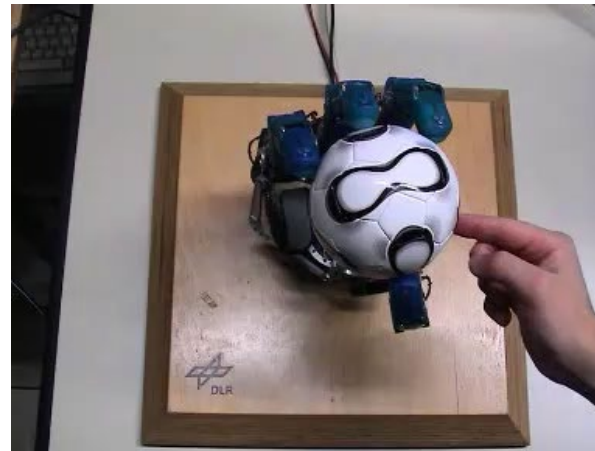
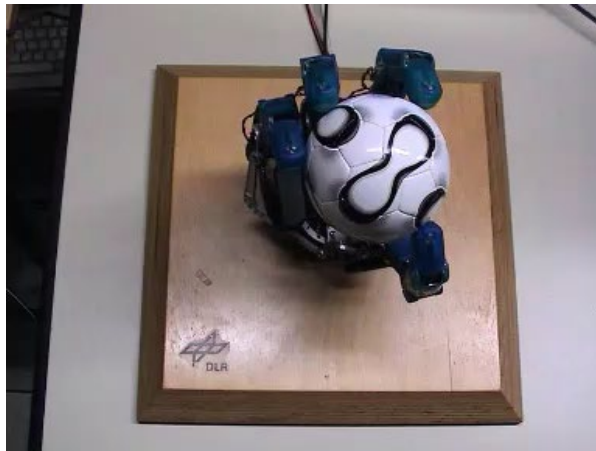


FIGURE 2.2: State of art dexterous tele-manipulation robotic system



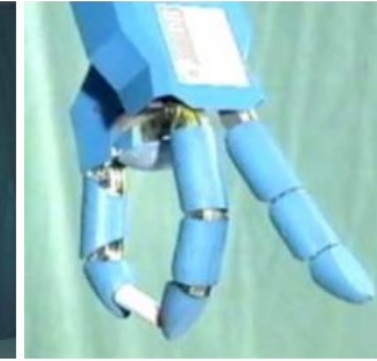
DLR hand – SOTA hand for many years



Robotic Hands



(a) UB Hand I



(b) UB Hand II

FIGURE 2.20: First two UB Hand prototypes

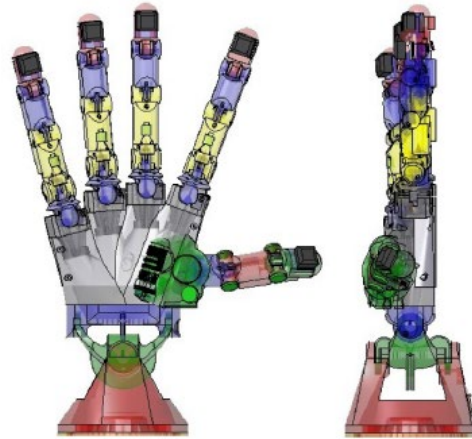
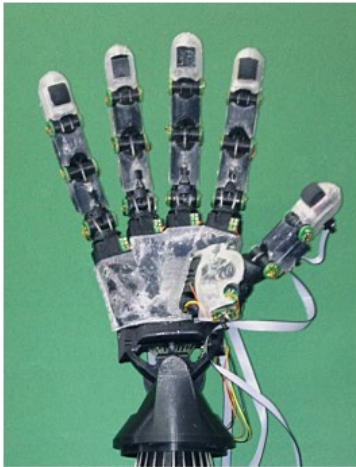
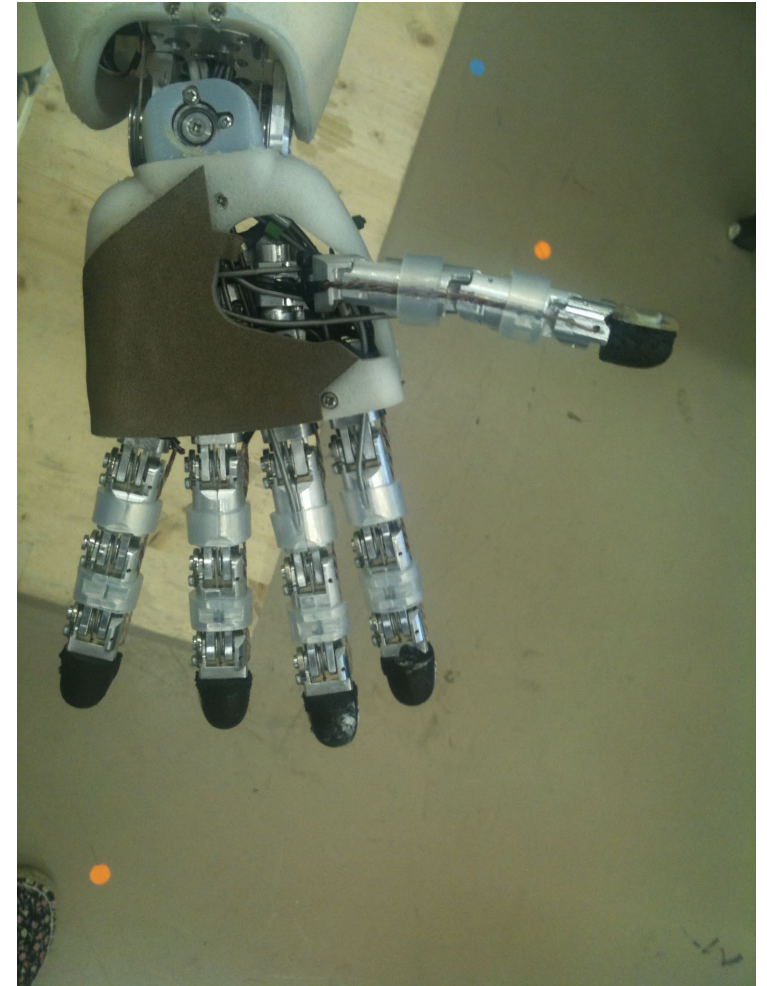


FIGURE 4.2: DEXMART Hand: Detailed view of the hand design.



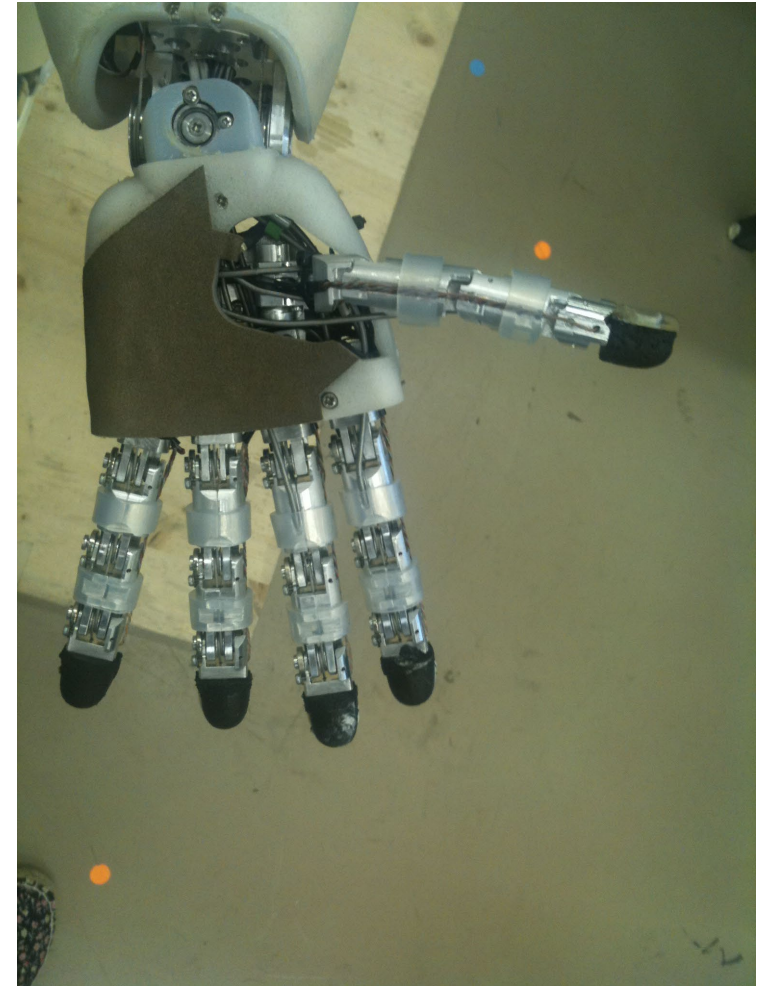
iCub Hand



Robotic Hands

Grasp Synthesis & Adaptation Learning

S. El Khoury, A.L. Pais, M. Li, and A.G. Billard
Learning Algorithms and Systems Laboratory (LASA)
Swiss Federal Institute of Technology (EPFL)



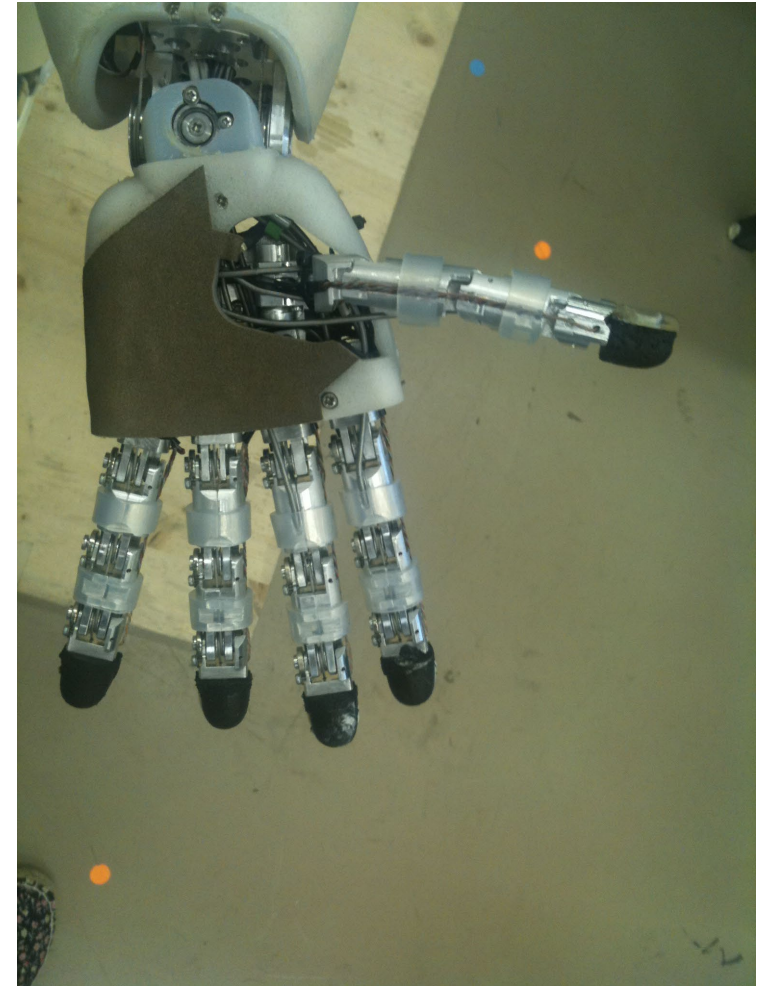
iCub Hand



Robotic Hands

Dexterous Bimanual Object Exploration with Whole-hand Tactile Sensing

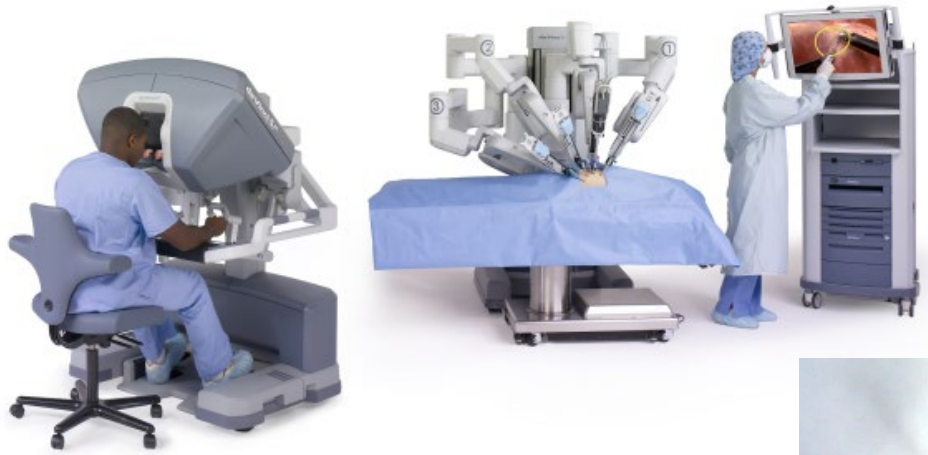
Nicolas Sommer, Miao Li
Aude Billard
Learning Algorithms and Systems Laboratory
EPFL



iCub Hand



Robotic Hands



(a) Da Vinci Robot



(b) Hand



(c) HMI

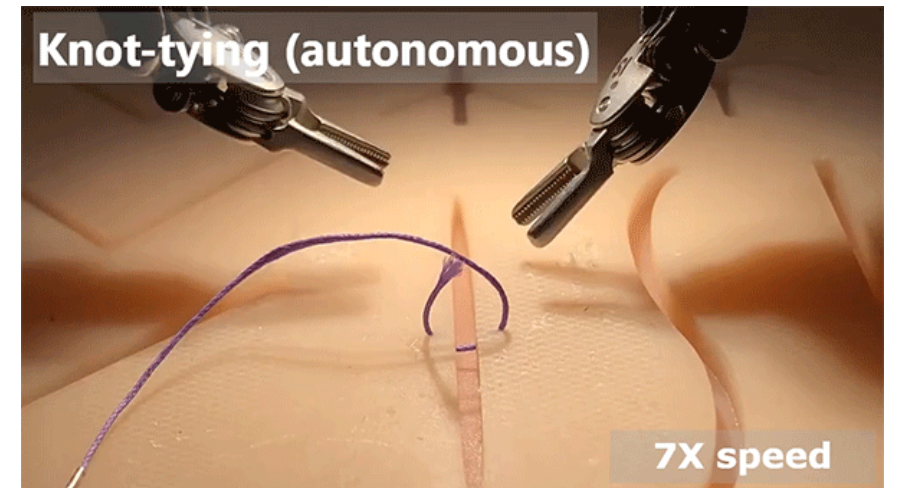
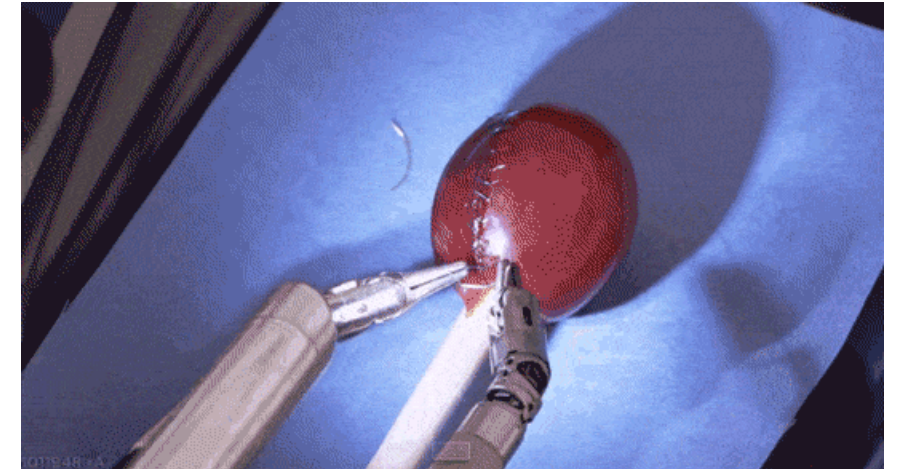


FIGURE 2.4: State of art telesurgery robotic system



Robotic Hands



(a) Toro (DLR)



(b) PR2 (WILLOW GARAGE)



(c) Asimo (HONDA)



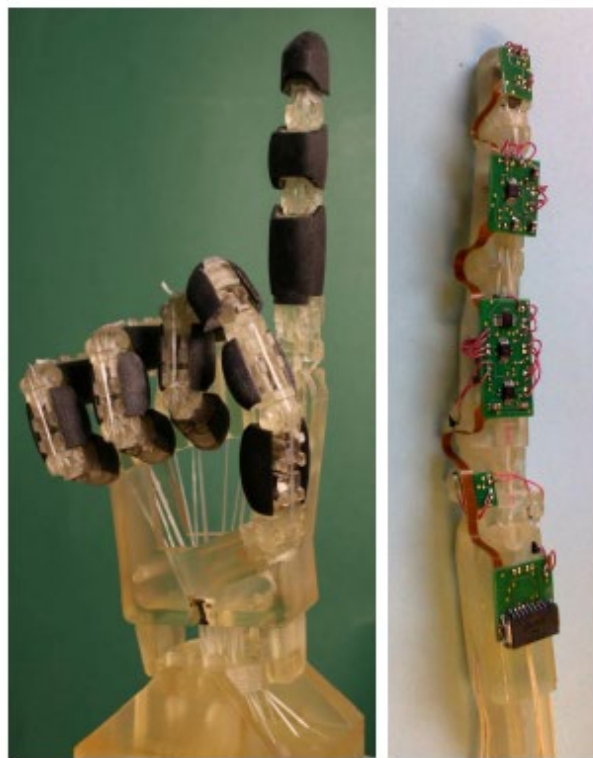
FIGURE 2.6: Humanoid robots



Robotic Hands



(a) UB Hand III



(b) UB Hand IV

FIGURE 2.21: Last two UB Hand prototypes

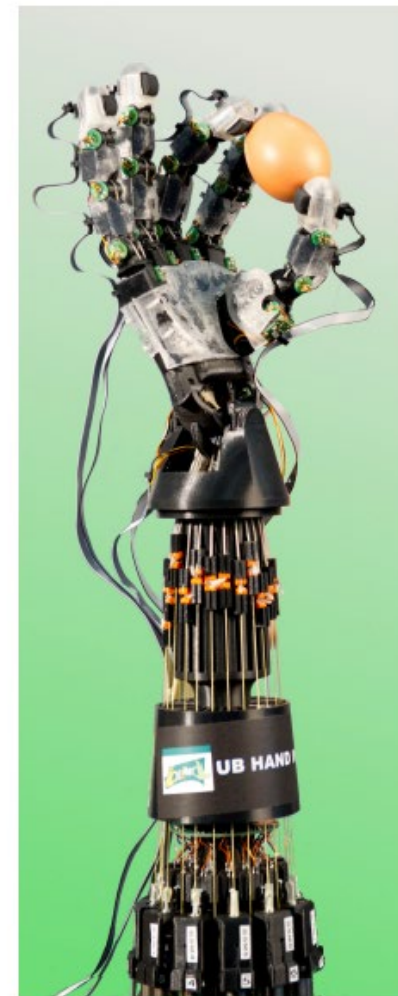


FIGURE 4.1: The DEXMART Hand.



Robotic Hands

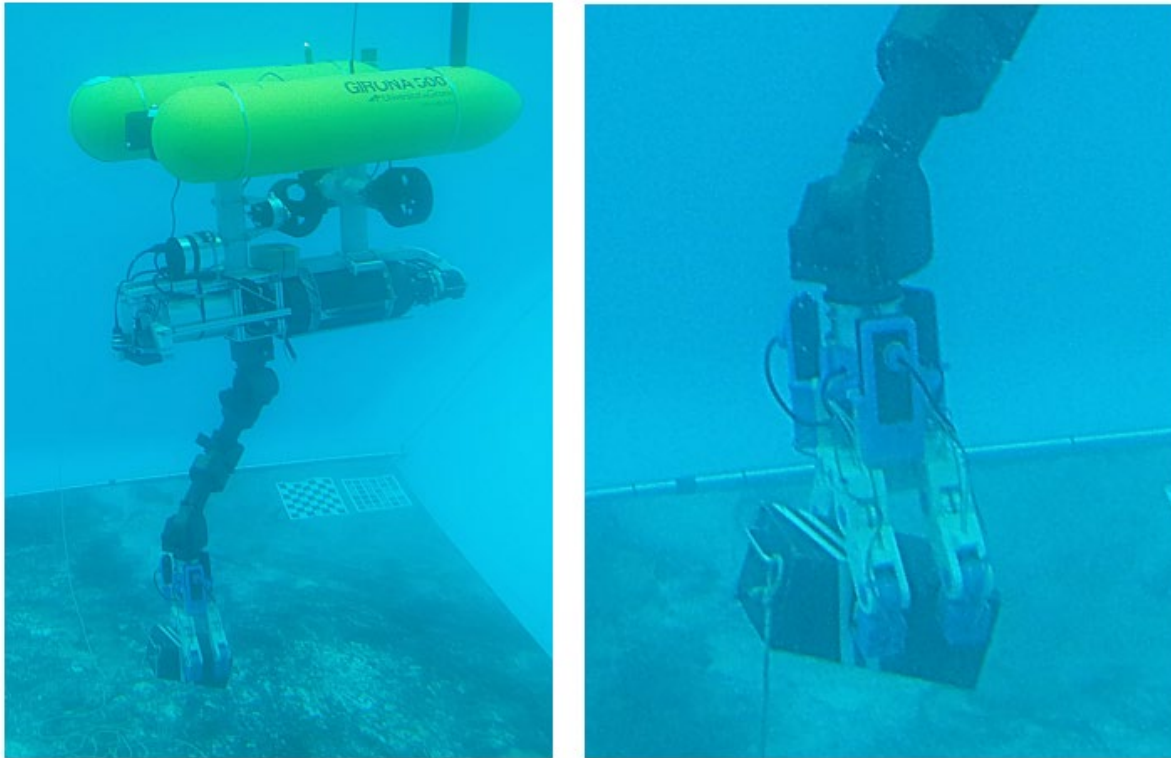
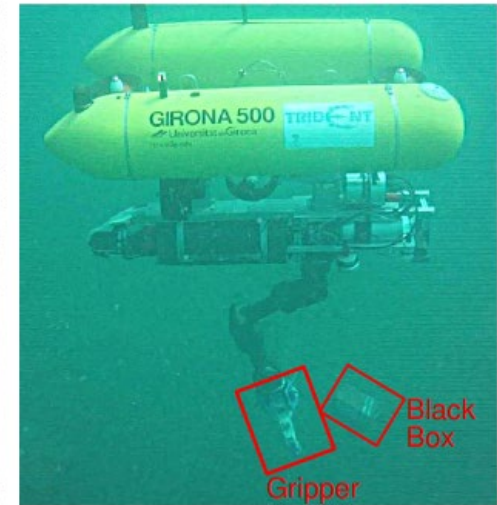


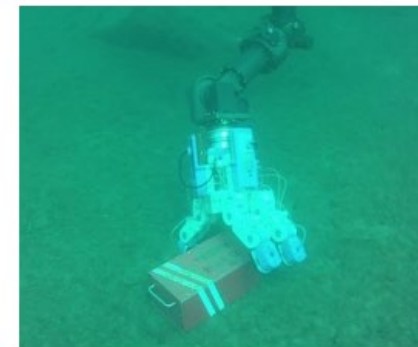
FIGURE 5.1: The Girona 500 AUV platform with the redundant arm and the gripper operating in the CIRS (Centre d'Investigació en Robòtica Submarina of the University of Girona) pool: a dummy black box is recovered from the pool floor.



(a) AUV deployment.



(b) AUV approaching the black box.



(c) The black box is approached.

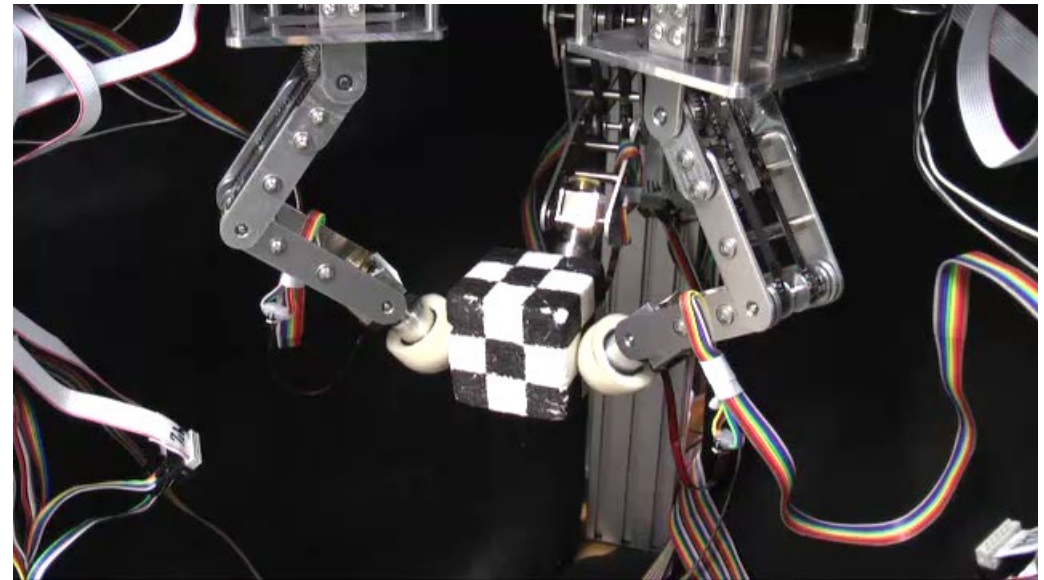
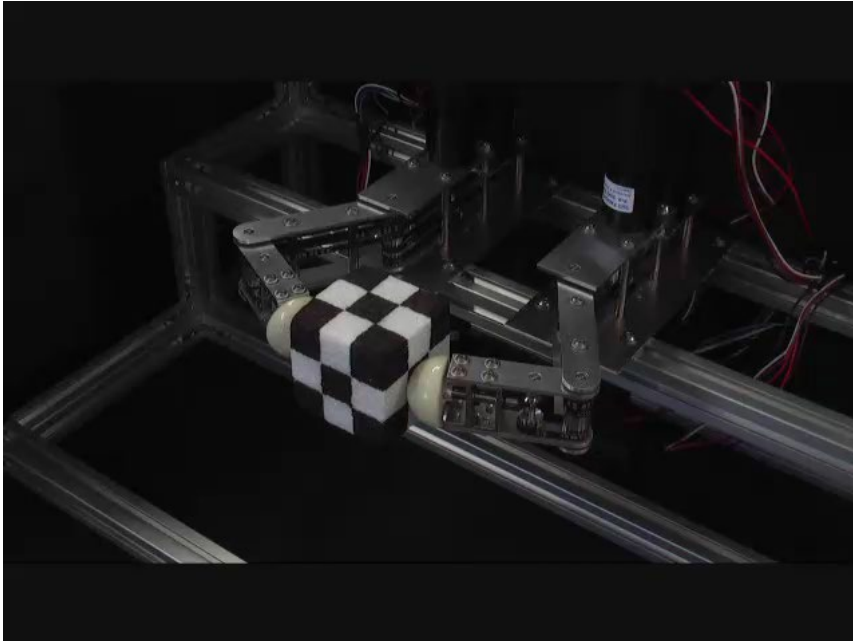
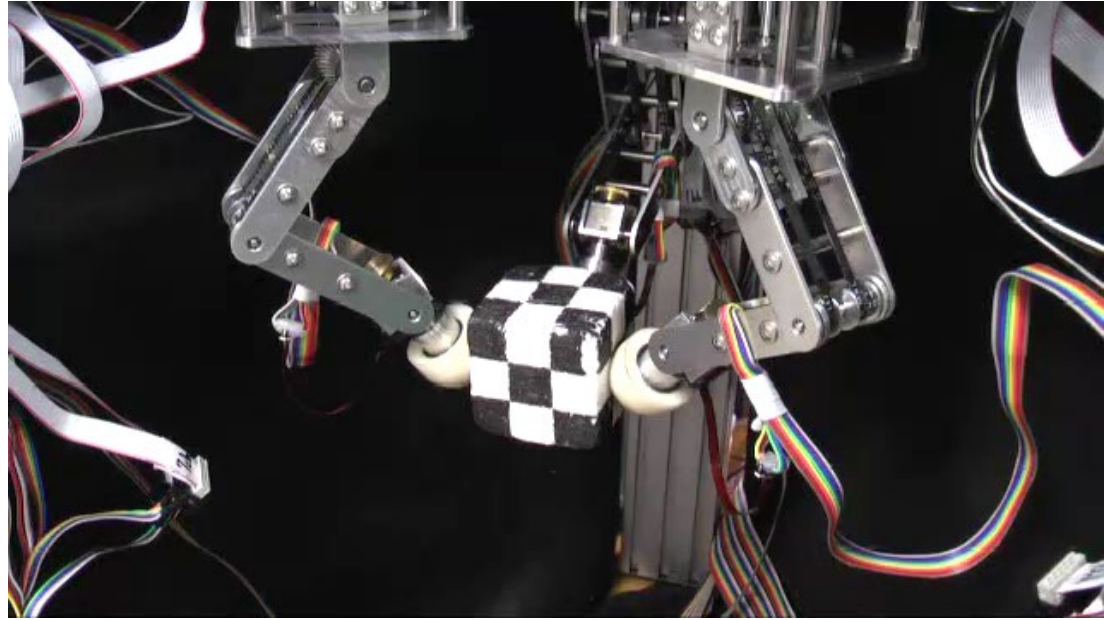
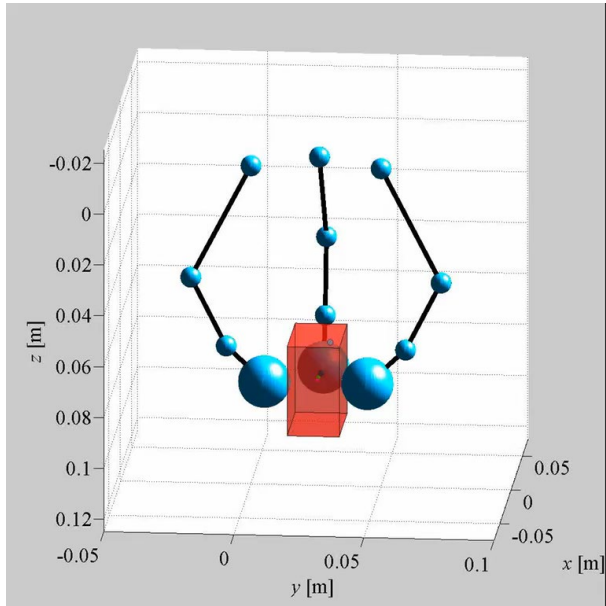


(d) Grasp execution seen from the AUV camera.

FIGURE 5.9: The AUV with integrated arm and gripper during the experimental tests at Port de Soller, Spain.



Robotic Hands





Robotic Hands

STABILITY OF VARIABLE GRASP STIFFNESS CONTROL

In this part, we address the control stability of varying stiffness in robotic grasping using a two-fingered examples, as shown in Fig. B.1(a). To this end, a detailed formulation for the dynamics of the hand-object system, including the rolling constraints and the soft fingertip constraints, is presented. The formulation on the grasping dynamics is based on the rigid body dynamics to grasp the object stably, which is derived from the proof of contact stability.

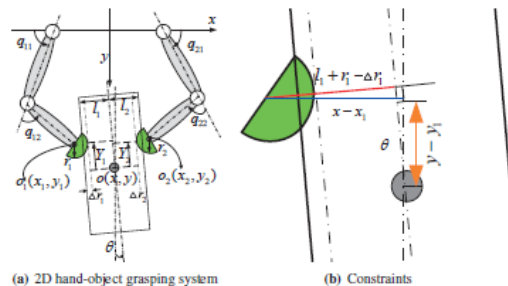


Figure B.1: (a) The hand-object system in 2D. Each finger has 2 DOFs with soft fingertips. (b) The constraint that the fingertip should keep contact with the object's surface.

B.1 DYNAMICS

In this part, we first consider the contact model of the fingertips and the constraints involved in the hand-object system. Then the overall dynamics of the system is formulated. The notations in this section are explained in Fig. B.1 to simplify the understanding of the grasping dynamics.

B.1.1 CONTACT MODEL OF SOFT FINGERTIP

$$f = c_1 (\Delta r)^2 + c_2 \frac{d}{dt} \Delta r \quad (\text{B.1})$$

where c_1 and c_2 are positive constant parameters which depend on the material of the fingertip, and Δr is the deformation at the fingertip. The fingertip should keep contact with the object surface, as shown in Fig. B.1(b), which can be expressed as follows

$$l_1 + r_1 - \Delta r_1 = (x - x_1) \cos \theta - (y - y_1) \sin \theta \quad (\text{B.2})$$

$$l_2 + r_2 - \Delta r_2 = -(x - x_2) \cos \theta + (y - y_2) \sin \theta \quad (\text{B.3})$$

B.1.2 ROLLING CONSTRAINTS

Complicated with many assumptions

$$Y_t = (x_t - x) \sin \theta + (y_t - y) \cos \theta \quad (\text{B.5})$$

$$q_{11} + q_{12} + \phi_1 = \pi + \theta \quad (\text{B.6})$$

$$q_{21} + q_{22} + \phi_2 = \pi - \theta \quad (\text{B.7})$$

B.1.3 OVERALL DYNAMICS

The total kinetic energy for the overall system can be described as follows

$$K = \sum_{i=1,2} \frac{1}{2} \dot{\mathbf{q}}_i^T H_i \dot{\mathbf{q}}_i + \frac{1}{2} M (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} I \dot{\theta}^2 \quad (\text{B.8})$$

where $\mathbf{q}_i = [q_{i1}, q_{i2}]^T$ is the vector of finger joints and $H_i \in \mathbb{R}^{2 \times 2}$ is the inertia matrix for each finger, M and I are the mass and inertia matrix of the object respectively.

The total potential energy from deformation can be given as:

$$P = \sum_{i=1,2} \int_0^{\Delta r_i} c_1 \eta^2 d\eta \quad (\text{B.9})$$

Then from the Hamilton's principle, we have

$$\int_{t_0}^{t_1} [\delta(K - P) - \sum_{i=1,2} c_2 \Delta \dot{r}_i \frac{\partial \Delta r_i}{\partial \dot{\mathbf{q}}_i^T} \delta \mathbf{X} + \sum_{i=1,2} \lambda_i [(r_i - \Delta r_i) \frac{\partial \phi_i}{\partial \mathbf{X}^T} + \frac{\partial Y_i}{\partial \mathbf{X}^T} \delta \mathbf{X} + \sum_{i=1,2} \mathbf{u}_i^T \delta \mathbf{q}_i] dt = 0 \quad (\text{B.10})$$

where $\mathbf{X} = [\mathbf{q}_1^T, \mathbf{q}_2^T, x, y, \theta]^T$.

Then we have the overall dynamics for the object-hand system as follows

$$H_t(\mathbf{q}) \ddot{\mathbf{q}}_t + (\frac{1}{2} \dot{H}_t + S_t) \dot{\mathbf{q}}_t + f_t \frac{\partial \Delta r_t}{\partial \dot{\mathbf{q}}_t^T} - \lambda_t [(r_t - \Delta r_t) \frac{\partial \phi_t}{\partial \dot{\mathbf{q}}_t^T} + \frac{\partial Y_t}{\partial \dot{\mathbf{q}}_t^T}] = \mathbf{u}_t \quad (\text{B.11})$$

$$M \ddot{x} + \sum_{i=1,2} [f_i \frac{\partial \Delta r_i}{\partial x} - \lambda_i \frac{\partial Y_i}{\partial x}] = 0 \quad (\text{B.12})$$

$$M \ddot{y} + \sum_{i=1,2} [f_i \frac{\partial \Delta r_i}{\partial y} - \lambda_i \frac{\partial Y_i}{\partial y}] = 0 \quad (\text{B.13})$$

$$I \ddot{\theta} + \sum_{i=1,2} [f_i \frac{\partial \Delta r_i}{\partial \theta} - \lambda_i ((r_i - \Delta r_i) \frac{\partial \phi_i}{\partial \theta} + \frac{\partial Y_i}{\partial \theta})] = 0 \quad (\text{B.14})$$

With the identities in section B.5, the overall dynamics can be simplified as

$$M \ddot{x} - (f_1 - f_2) \cos \theta + (\lambda_1 + \lambda_2) \sin \theta = 0 \quad (\text{B.15})$$

$$M \ddot{y} + (f_1 - f_2) \sin \theta + (\lambda_1 + \lambda_2) \cos \theta = 0 \quad (\text{B.16})$$

$$I \ddot{\theta} - f_1 Y_1 + f_2 Y_2 + l_1 \lambda_1 - l_2 \lambda_2 = 0 \quad (\text{B.17})$$

$$I \ddot{\theta} - f_1 Y_1 + f_2 Y_2 + l_1 \lambda_1 - l_2 \lambda_2 = 0 \quad (\text{B.18})$$

B.2 VARIABLE GRASP STIFFNESS CONTROL

Motivated by the analysis of the overall system dynamics, the following control law is adopted for each finger to achieve stable grasp

$$\mathbf{u}_i = -D_i \dot{\mathbf{q}}_i + k J_i^T (\mathbf{x}_i - \mathbf{x}_d) \quad (\text{B.19})$$

$$\mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \quad \mathbf{x}_d = \frac{1}{2} \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \quad (\text{B.20})$$

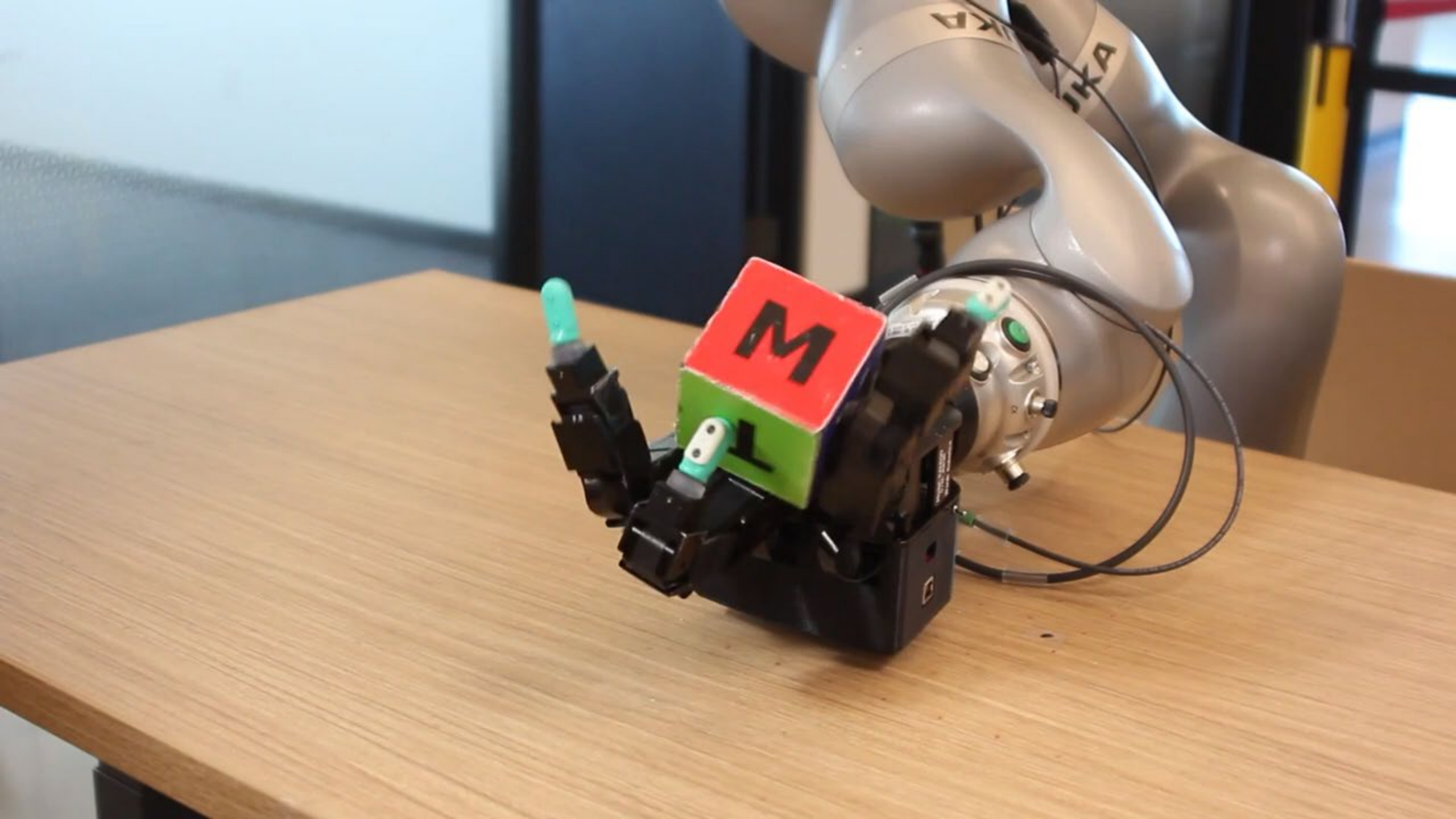
where D_i is a diagonal positive definite matrix representing the damping gain. $k \in \mathbb{R}^+$ represents the variable stiffness for each fingertip (k is the same value for the two-finger grasp to ensure force balance).

B.3 STABILITY PROOF-1

Taking the sum of inner product of Eq. (B.15) with $\dot{\mathbf{q}}_1$, $i = 1, 2$, Eq. (B.16) with $\dot{x}$, Eq. (B.17) with $\dot{y}$, Eq. (B.18) with $\dot{\theta}$, we have

$$\frac{d}{dt} E = - \sum_{i=1,2} \{\dot{\mathbf{q}}_i^T D_i \dot{\mathbf{q}}_i + c_2 \Delta \dot{r}_i^2\} + \frac{k}{4} (\mathbf{x}_1 - \mathbf{x}_2)^T (\mathbf{x}_1 - \mathbf{x}_2) \quad (\text{B.21})$$

$$E = K + V + P \quad (\text{B.22})$$





Opening thoughts on robot hands

- We have had high degree of freedom robot hands in humanlike form since the 80's
- There have been many exciting new ideas about hand design throughout the past decades.
- Yet we still do not have highly dexterous robots
- Why is this the case?
- What are the gaps?
- How can we close them?



Opening thoughts on robot hands

JANUARY 30, 2017 — QUICK TAKES

Research Needed on Robot Hands

rodneymrooks.com/research-needed-on-robot-hands/

This is a short piece I wrote for a workshop on what are good things to work on in robotics research.

My strawman is that we will need concurrent progress in at least five areas, each feeding off the other, in order to come up with truly useful and general robot hands:

- new (low cost) mechanisms for both kinematics and force control
- materials to act as a skin (grasp properties and longevity)
- long life sensors that can be embedded in the skin and mechanism
- algorithms to dynamically adjust grasps based on sensing
- learning mechanisms on visual/3D data to inform hands for pregrasp

I think progress on one of these alone is hard to get adopted by research groups working on others. The constraints between them are not well understood and need to be challenged and adapted to by all the researchers. This is a tall order. This is why grippers on factory robots today look just like they did forty years ago.

“Human hands just work. Give them an object from a very wide class of objects and they grip that object, usually with a wide variety of possible grips. They sense when the grip is failing and adjust. **They work reliably and quickly.**”



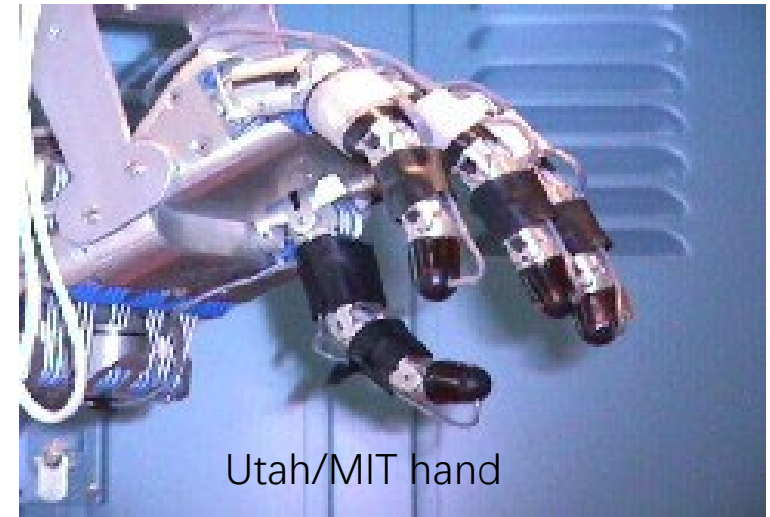
Opening thoughts on robot hands

- Complex rigid hands = Complicated!

Difficult to control

Expensive (? Not today)

Fragile



Utah/MIT hand

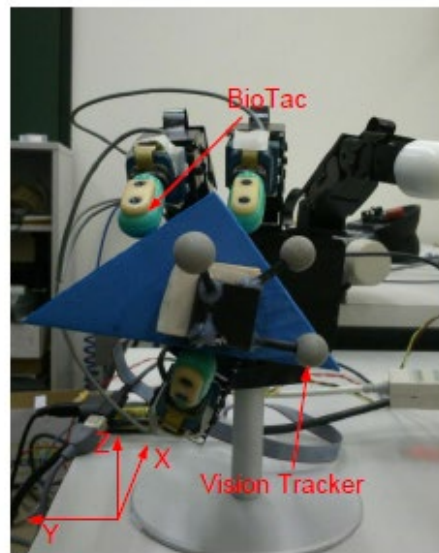
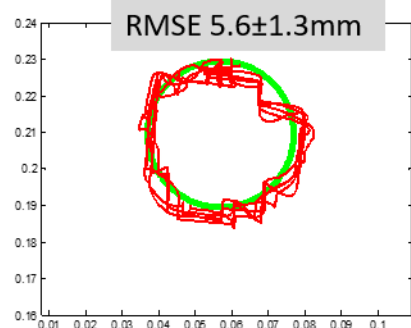
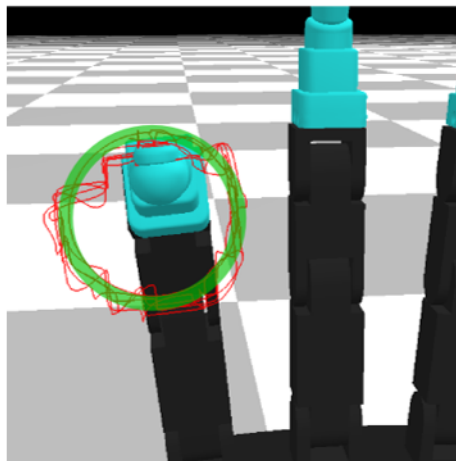
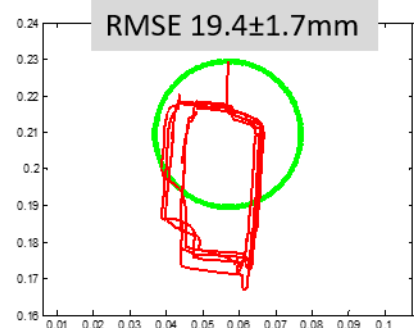
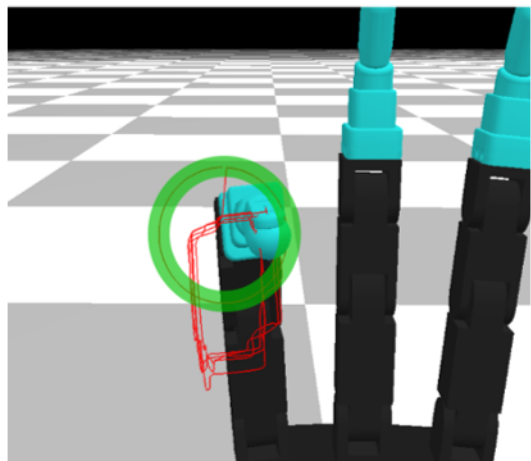
robonaut.jsc.nasa.gov

They don't work reliably!



Opening thoughts on robot hands

Index



(a)

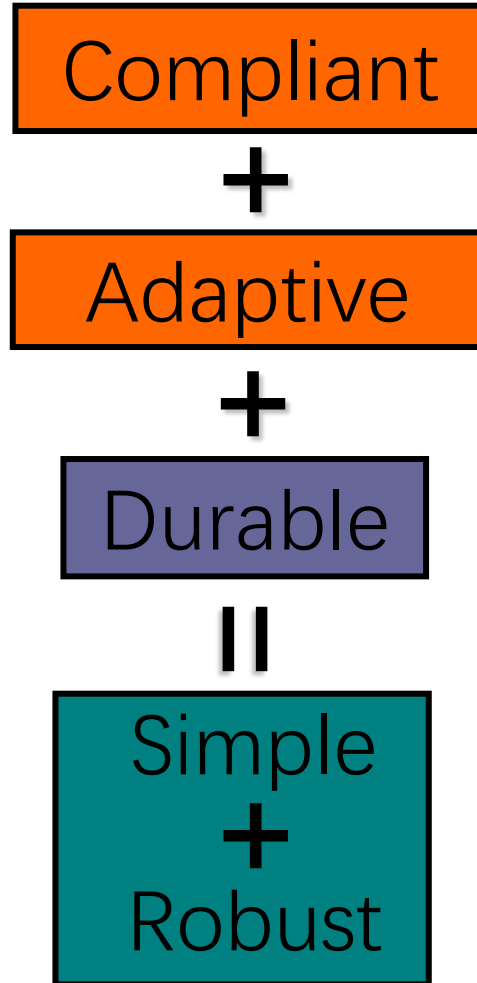


- How to deal with “poor” sensing?
 - Errors in positioning, finger placement
 - Can't control contact forces

Grasp will likely be unsuccessful



Open Hand Project



Aaron M. Dollar

Professor of Mechanical Engineering & Materials Science and Computer Science,
[Yale University](#)
Verified email at yale.edu - [Homepage](#)
[Robotics](#) [Mechanisms](#) [Biomechanics](#) [Manipulation](#) [Rehabilitation Robotics](#)

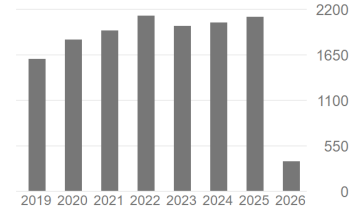
FOLLOWING

TITLE	CITED BY	YEAR
Lower extremity exoskeletons and active orthoses: challenges and state-of-the-art AM Dollar, H Herr IEEE Transactions on robotics 24 (1), 144-158	1971	2008
The grasp taxonomy of human grasp types T Feix, J Romero, HB Schmiedmayer, AM Dollar, D Kragic IEEE Transactions on human-machine systems 46 (1), 66-77	1403	2015
The ycb object and model set: Towards common benchmarks for manipulation research B Calli, A Singh, A Walsman, S Srinivasa, P Abbeel, AM Dollar 2015 international conference on advanced robotics (ICAR), 510-517	1233	2015
Benchmarking in manipulation research: Using the Yale-CMU-Berkeley object and model set B Calli, A Walsman, A Singh, S Srinivasa, P Abbeel, AM Dollar IEEE Robotics & Automation Magazine 22 (3), 36-52	1083 *	2015

Cited by

[VIEW ALL](#)

	All	Since 2021
Citations	21165	10610
h-index	67	49
i10-index	186	136



Public access

[VIEW ALL](#)

5 articles

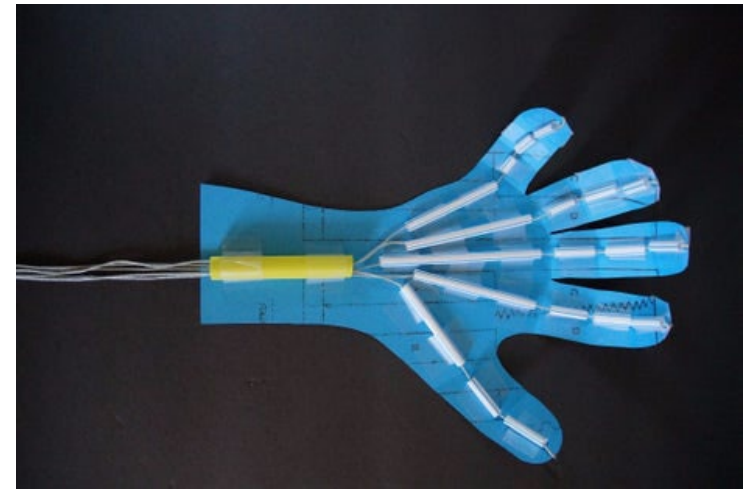
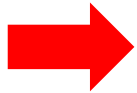
82 articles

“Smart” **mechanical design** for simplicity of use and robust manipulation

- 3D printing
- Soft hand technology
- Anyone can make a hand?



A toy example





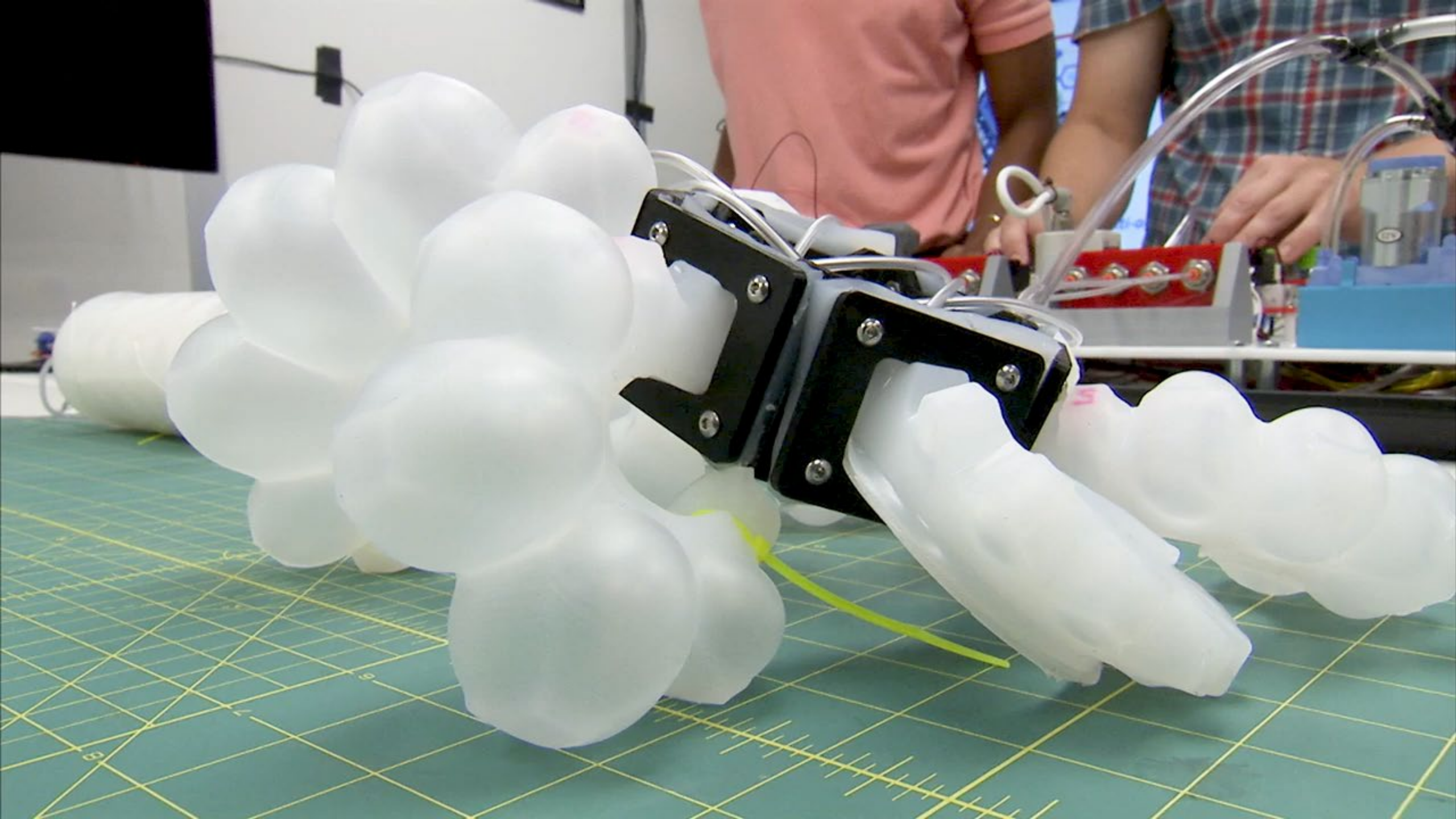
Video
(2 mins)

Video
(7 mins)

Soft Robotics

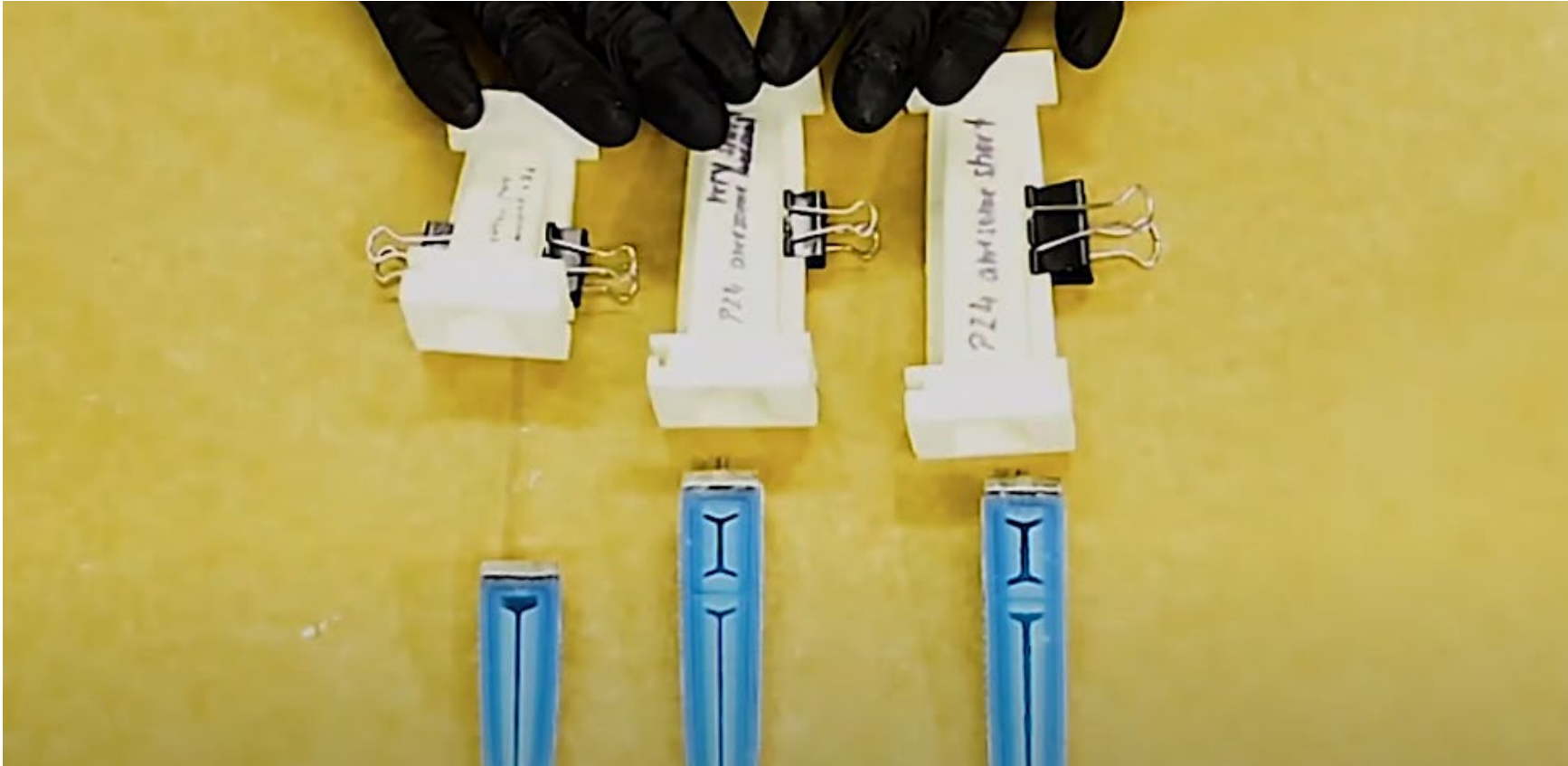
Soft robotic gripper







Soft Hands



How to build Soft Robotics Fingers, Pulps and Gloves

https://www.youtube.com/watch?v=HE4MGYLkXjk&ab_channel=RBOTUBerlin



Soft Hands

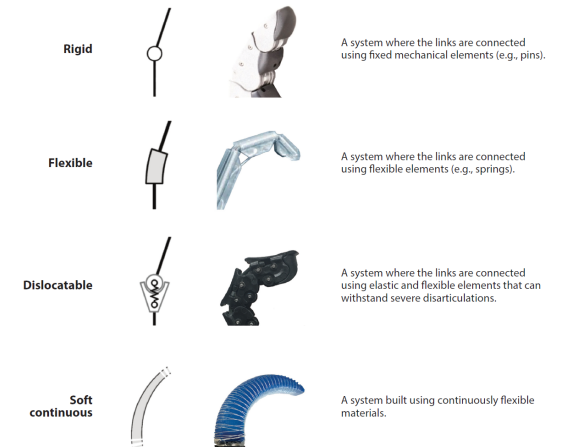
Rigid Hand



Soft Hand



Deformation



How to design and control the deformation?



Soft Hands

Rigid



A system where the links are connected using fixed mechanical elements (e.g., pins).



Flexible



A system where the links are connected using flexible elements (e.g., springs).



Dislocatable



A system where the links are connected using elastic and flexible elements that can withstand severe disarticulations.



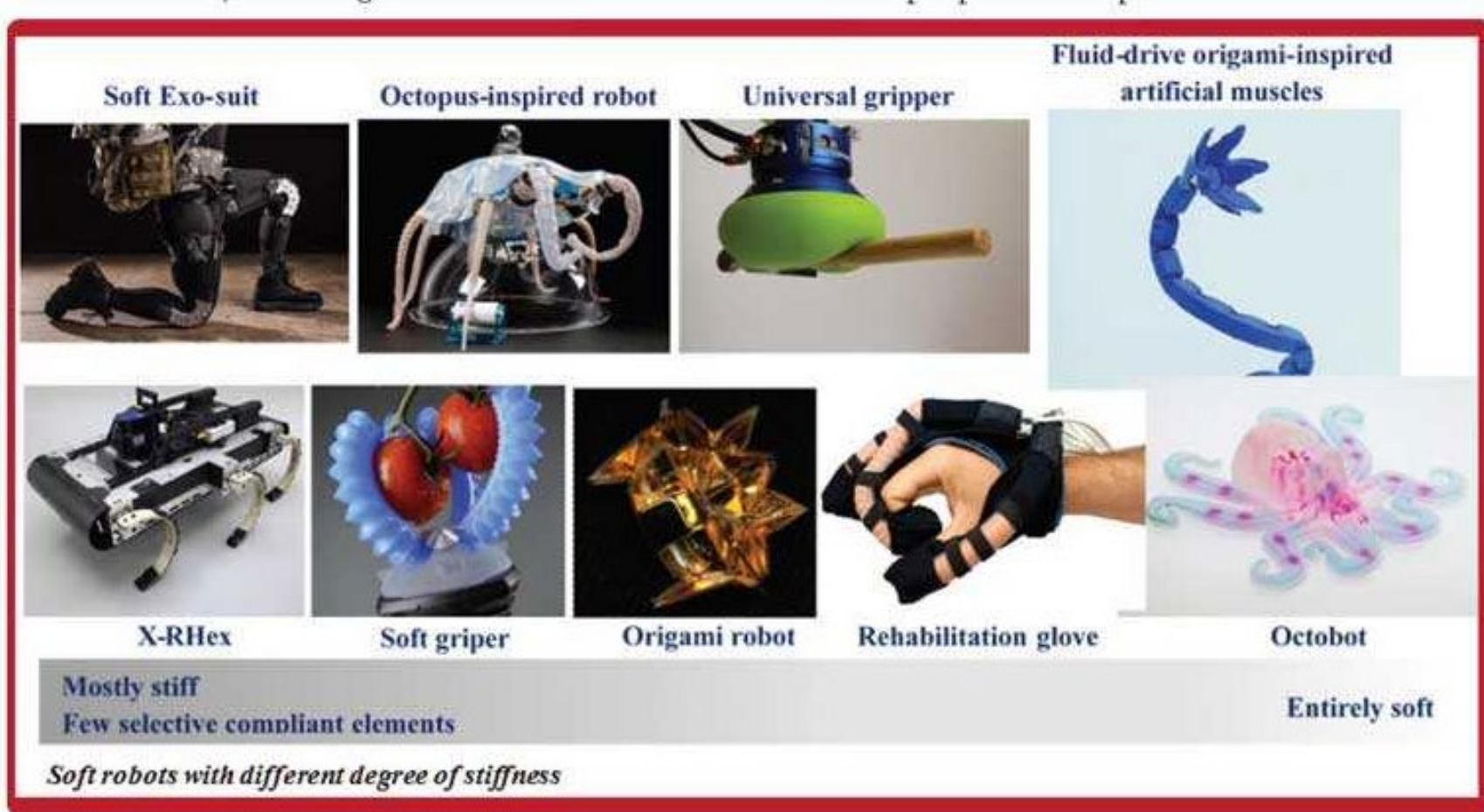
Soft continuous



A system built using continuously flexible materials.



Soft Hands



How to design and control the deformation?

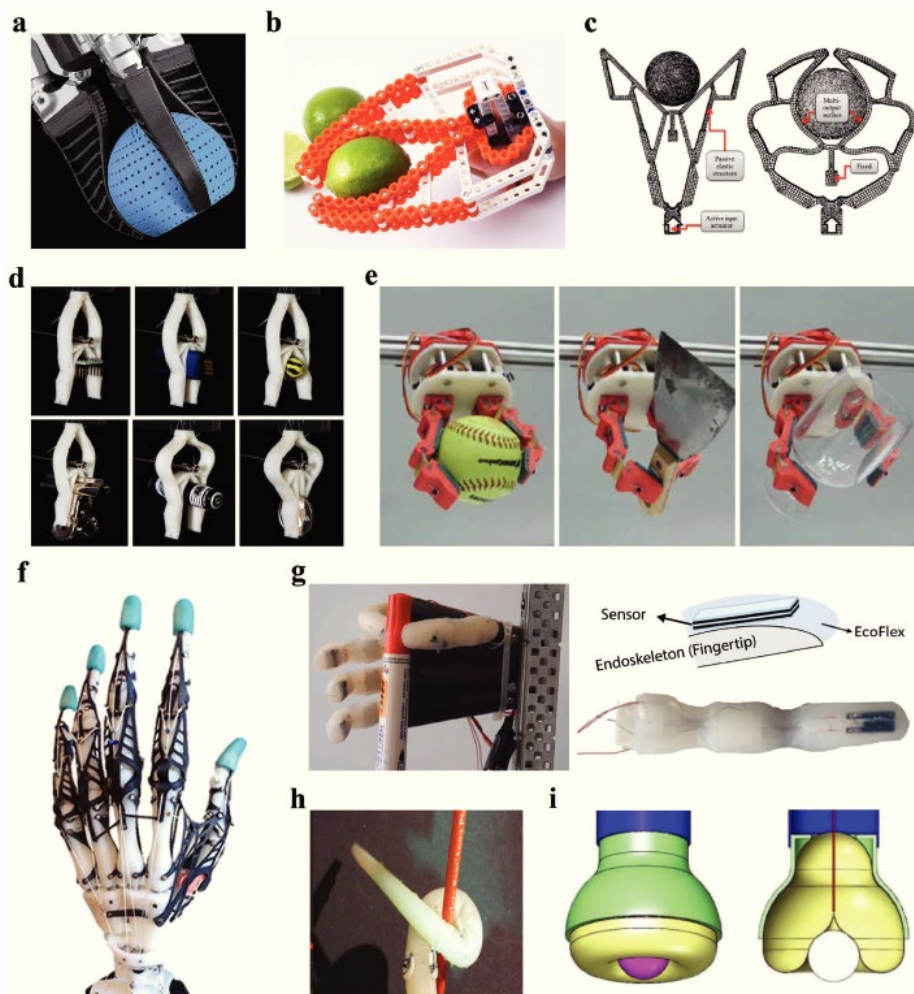
Soft Hands

REVIEW

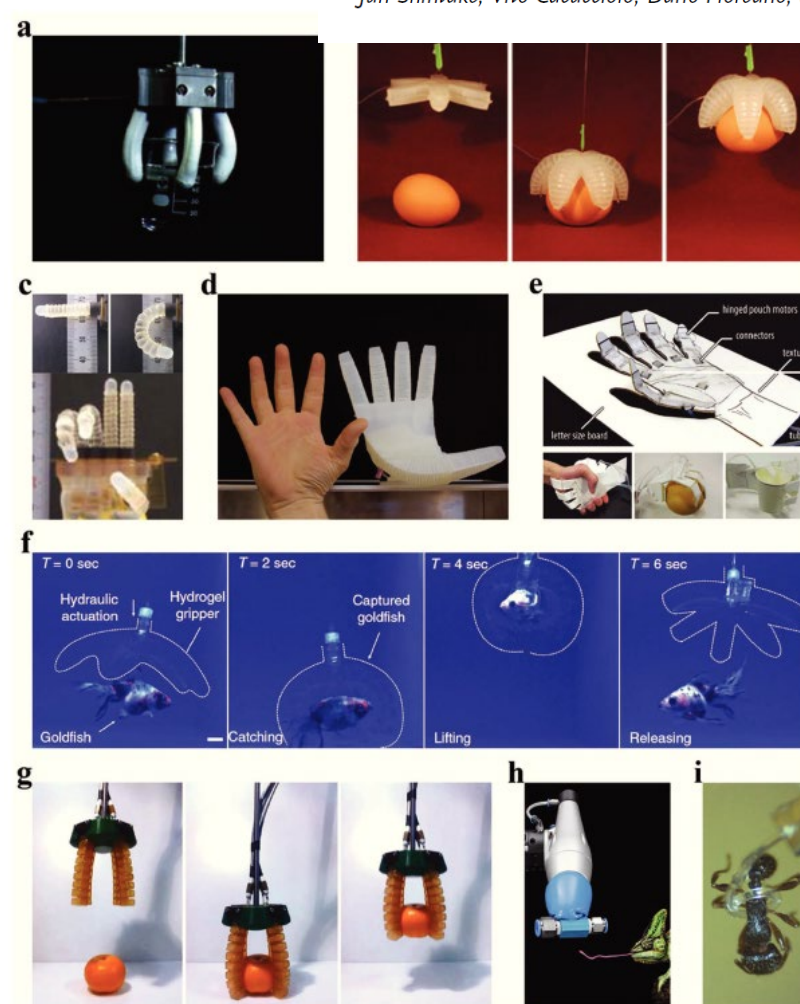
Soft Grippers

Soft Robotic Grippers

*Jun Shintake, Vito Cacucciolo, Dario Floreano, and Herbert Shea**



Soft grippers using passive structure with external motors



Soft grippers using fluidic elastomer actuators

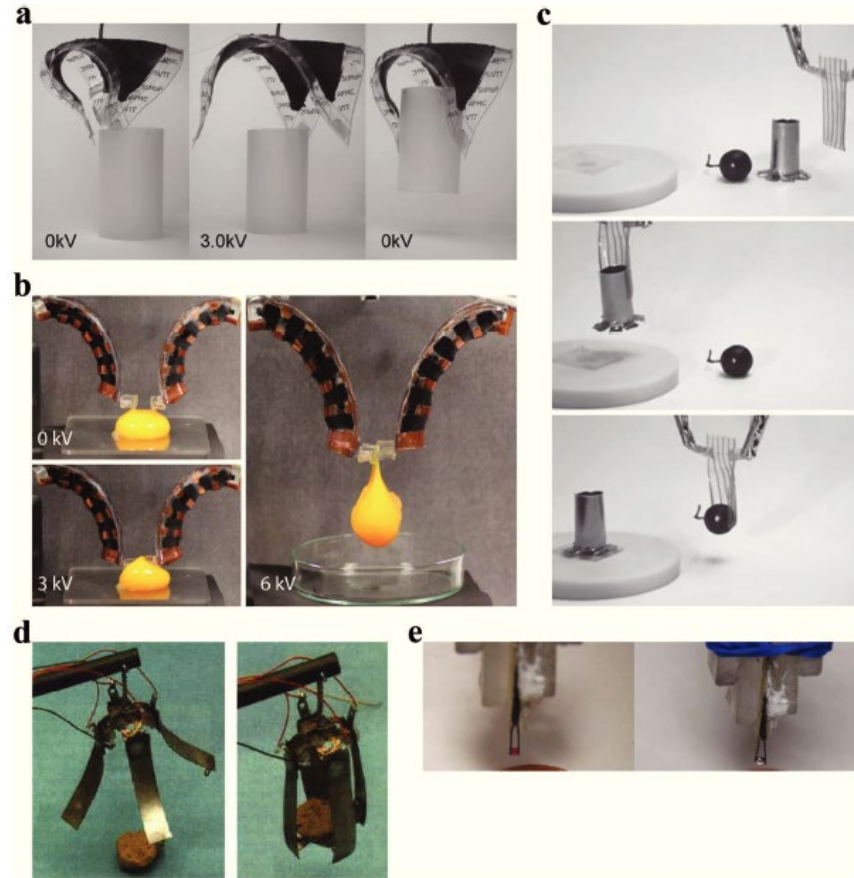
Soft Hands

REVIEW

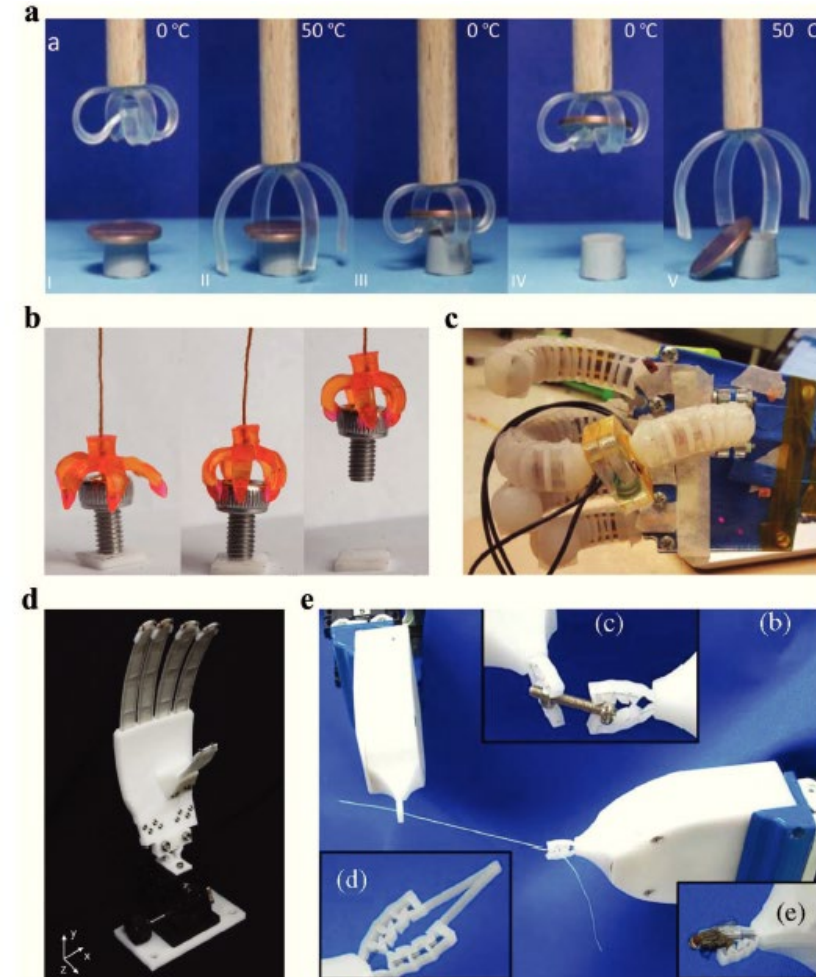
Soft Grippers

Soft Robotic Grippers

*Jun Shintake, Vito Cacucciolo, Dario Floreano, and Herbert Shea**



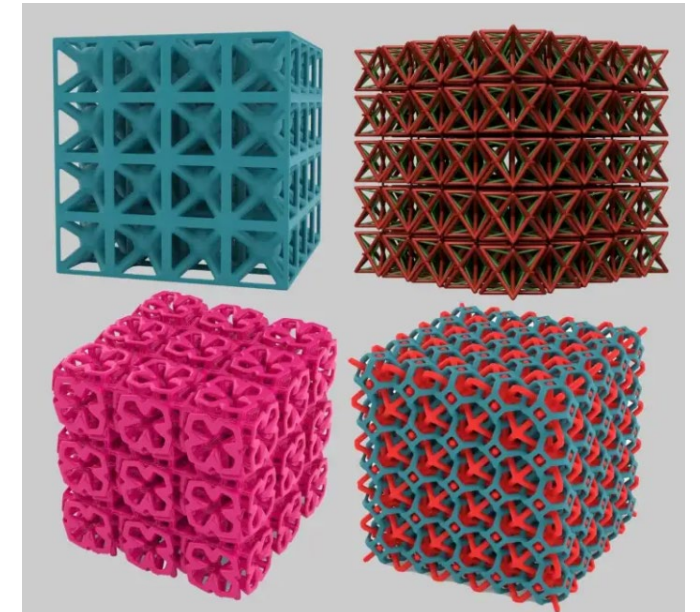
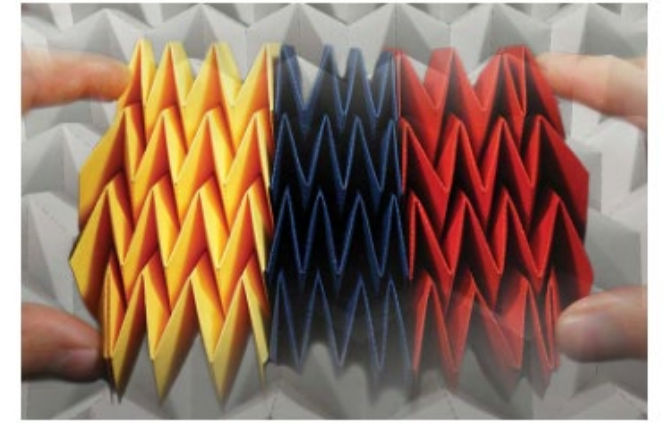
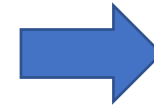
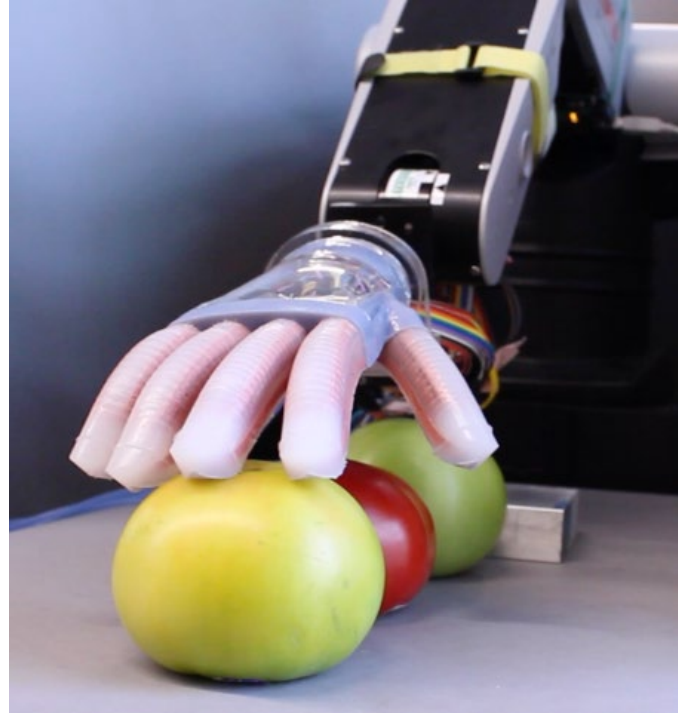
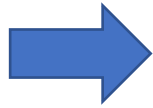
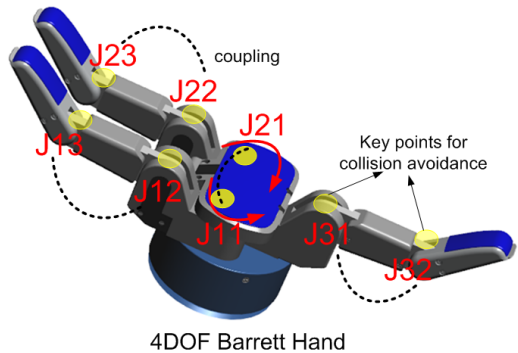
Soft grippers using dielectric elastomer actuators



Soft grippers using shape memory materials



New Material



Steel → Rubber....→ Meta Material

Video
(3 mins)

Grasping without Squeezing: Shear Adhesion Gripper with Fibrillar Thin Film

E.W. Hawkes, D.L. Christensen, A.K. Han, H. Jiang, and M.R. Cutkosky
Stanford University

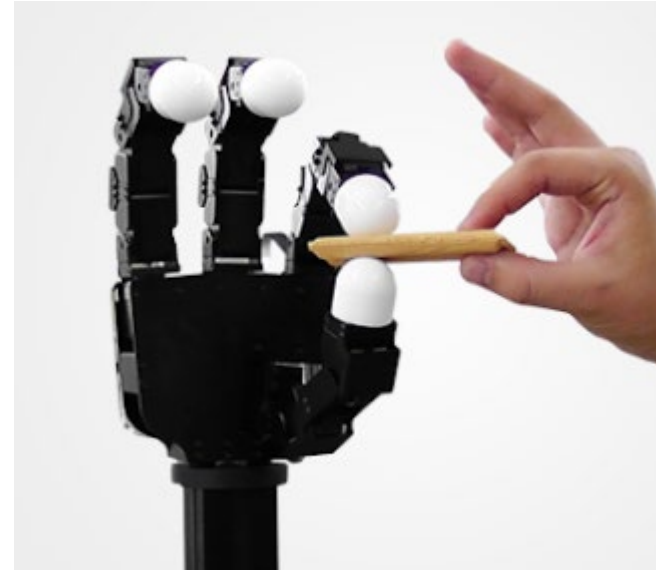
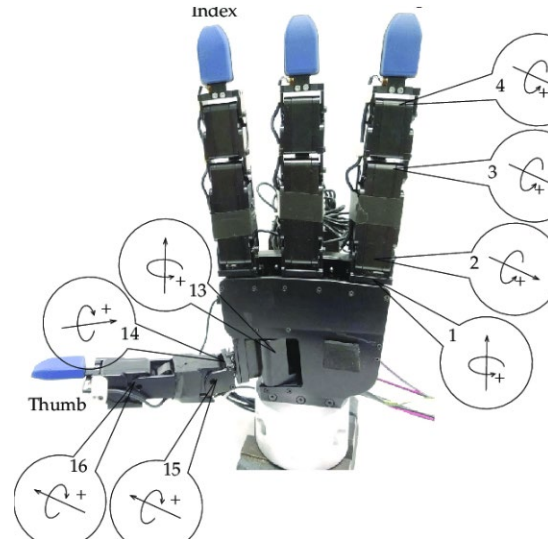


Design, Fabrication, and Evaluation of Tendon-Driven Foam Manipulators

Jonathan P. King, Dominik Bauer, Cornelia Schlagenhauf,
Kai-Hung Chang, Daniele Moro, Nancy Pollard, and Stelian Coros



Motion Transmission

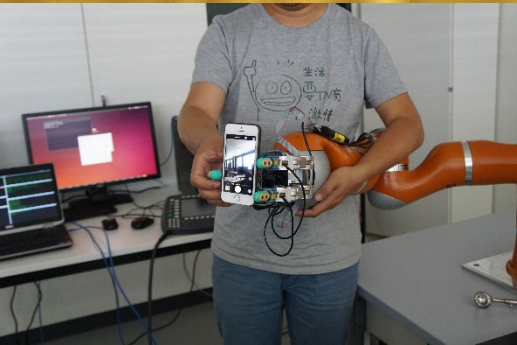
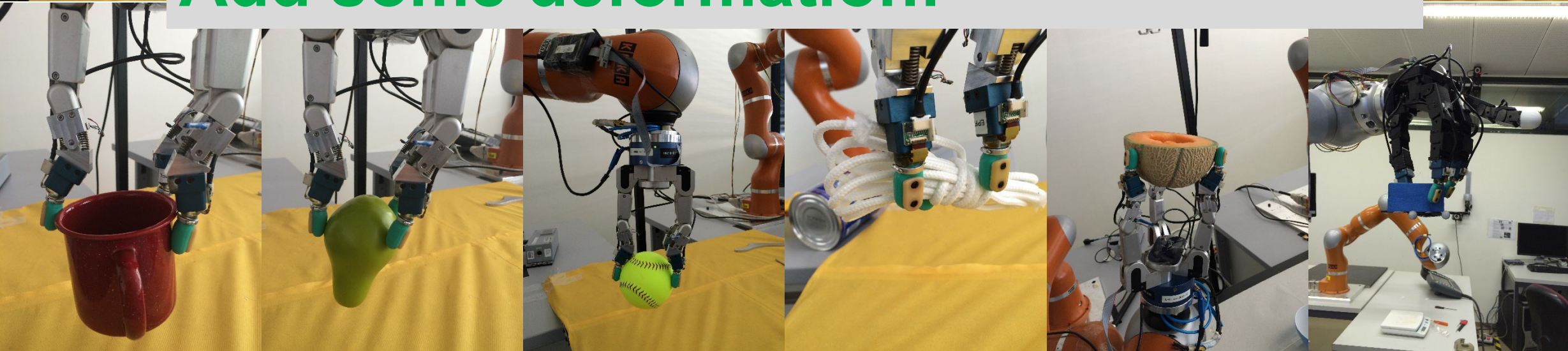


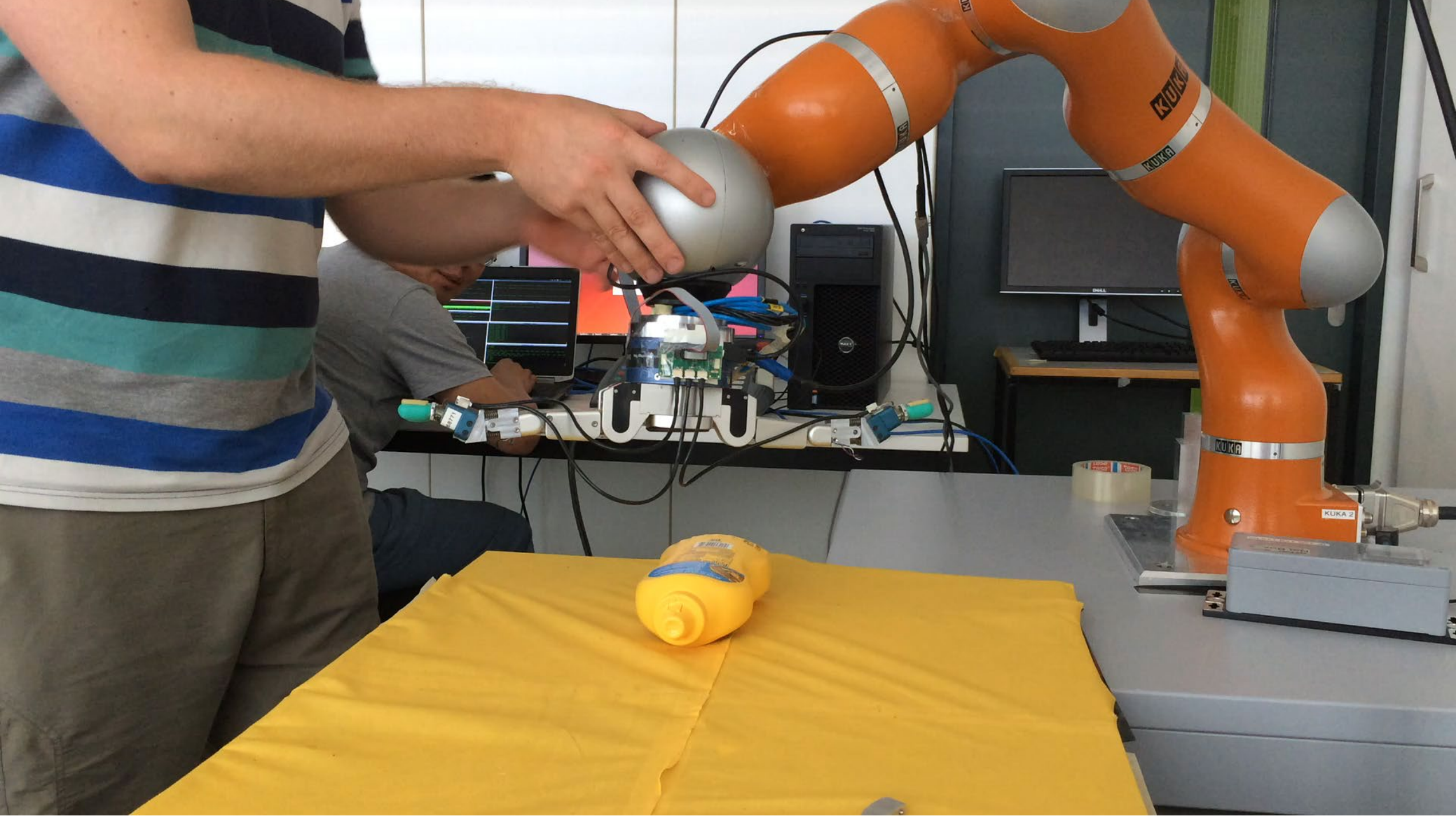
No(few) Deformation!

**Backdrivable is already good enough
for rigid hand!**



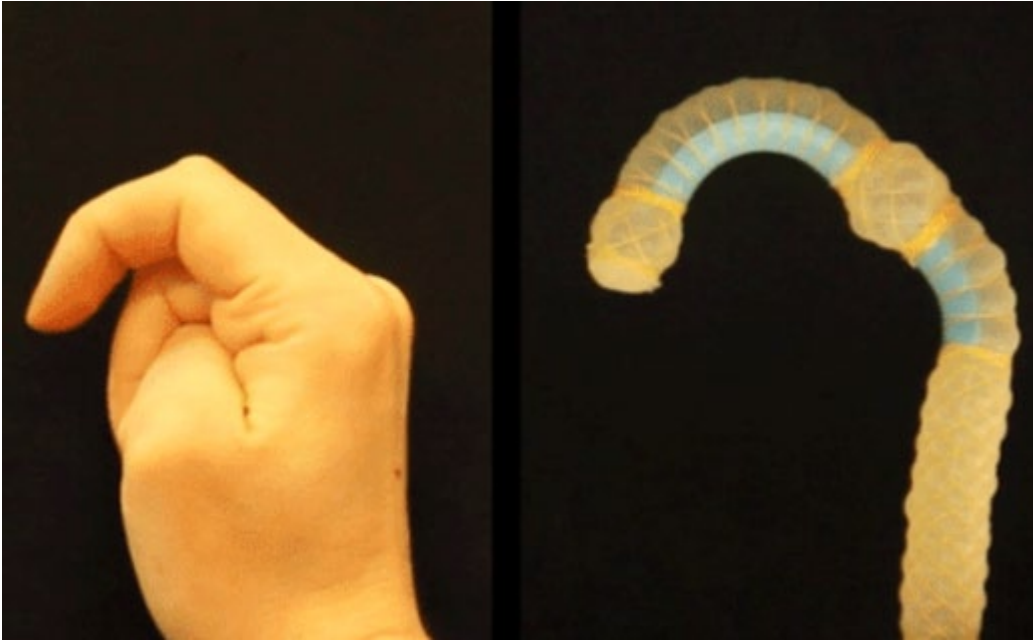
Add some deformation!







Motion Transmission



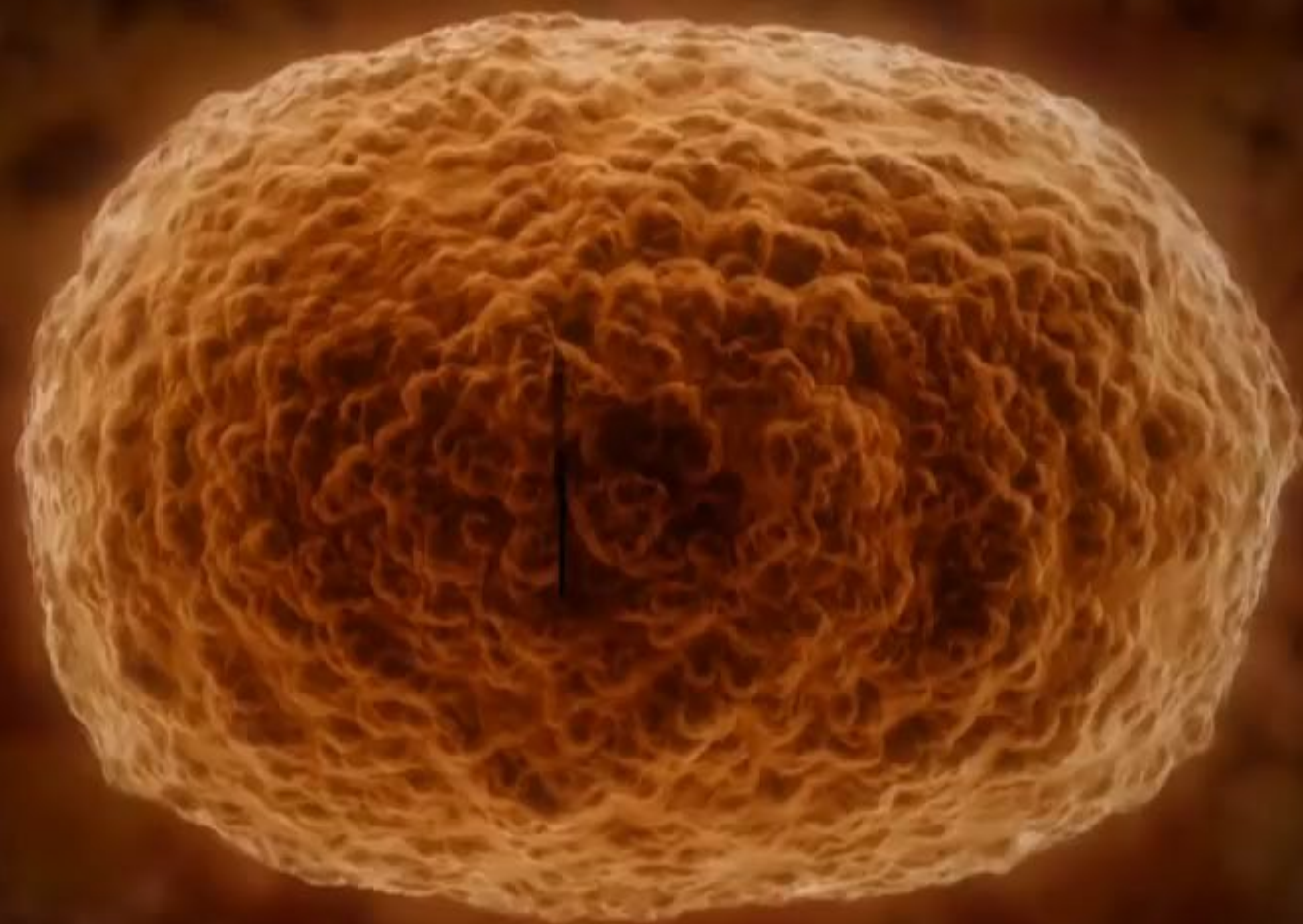
Deformation is motion!
Shape is function!

Video
(1 mins)

HOW DOES A ROBOT PICK UP A RASPBERRY WITHOUT SQUISHING IT?

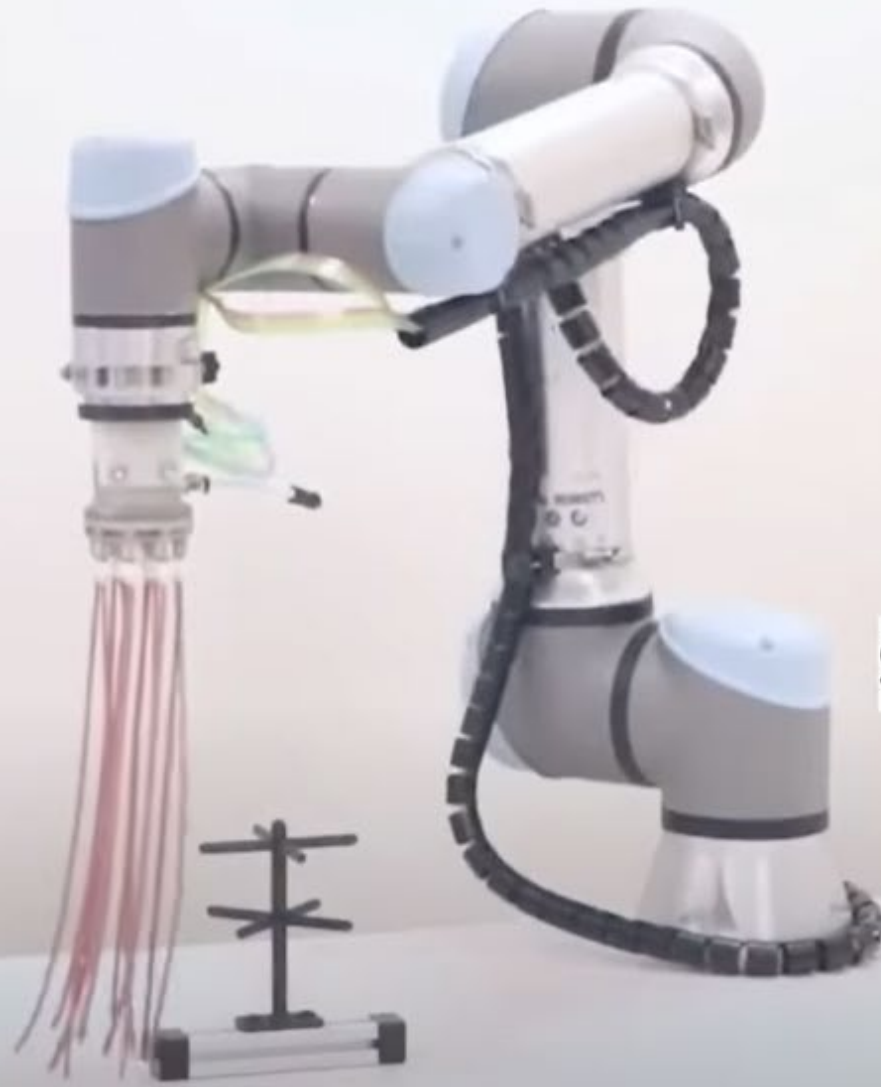


Video
(2 mins)



Video
(1 mins)

**But this gripper
relies on simple
inflation to
entangle objects
and doesn't
require any
sensing, planning,
or feedback
control.**



**Video
(1 mins)**





Actuator

- electric motors;
- pneumatic actuators;
- hydraulic actuators;
- shape memory alloys (SMA).

Proprieties class	Power density ρ [W/Kg]	$\sigma_{\max}$ [MPa]	$\epsilon_{\max}$	E [GPa]	Efficiency
DC motors	100	0.1	0.4	*	0.6–0.8
Pneumatic	400	0.5–0.9	1	$5-9 \times 10^{-4}$	0.4–0.5
Hydraulic	2,000	20–70	1	2–3	0.9–0.98
SMA	1,000	100–700	0.07	30–90	0.01–0.02
Human muscle	500	0.1–0.4	0.3–0.7	0.005–0.09	0.2–0.25

FIGURE 2.11: Actuator Performance Indices: Power density ρ = Power per unit of weight, $\sigma_{\max}$ = Maximum force exerted by the actuators per area, $\epsilon_{\max}$ = Maximum run per length, E Actuator stiffness. Maximum stress and strain are indexes specifically designed for linear actuators. Units are expressed as follow: W Watt, Kg kilogram, MPa Mega Pascal, GPa Giga Pascal.

*Depending on the gearhead



Actuator



(a) McKibben actuators



(b) Shadow Hand finger set-up

FIGURE 2.12: Pneumatic actuators for robotic hands

Video
(5 mins)





Actuator

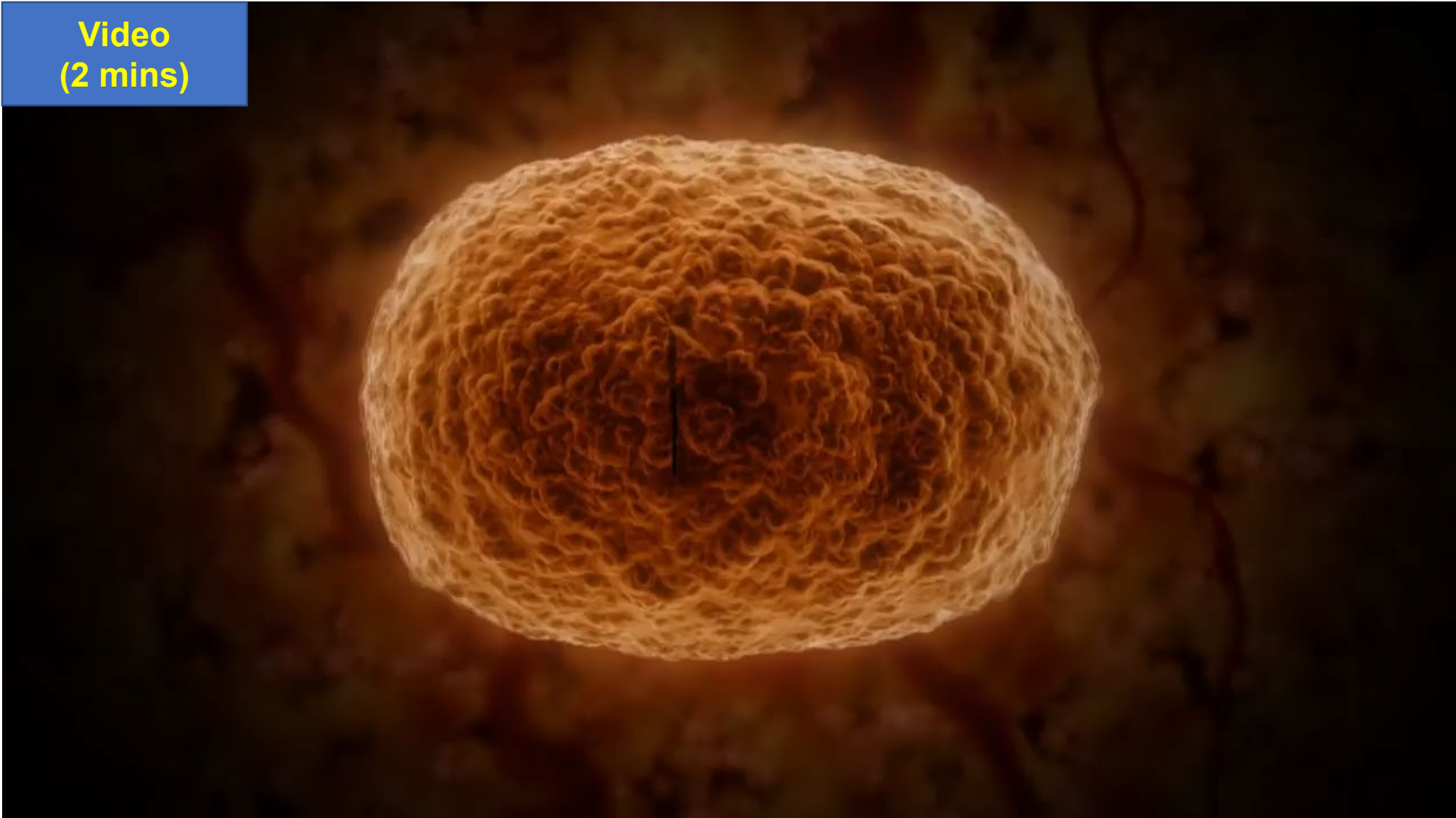
Video
(3 mins)





Actuator

Video
(2 mins)





Sensors

- motor position sensors;
- finger joints position sensors;
- motor torque sensors;
- joint torque sensors;
- tactile sensors;
- temperature sensors;
- in hand camera.

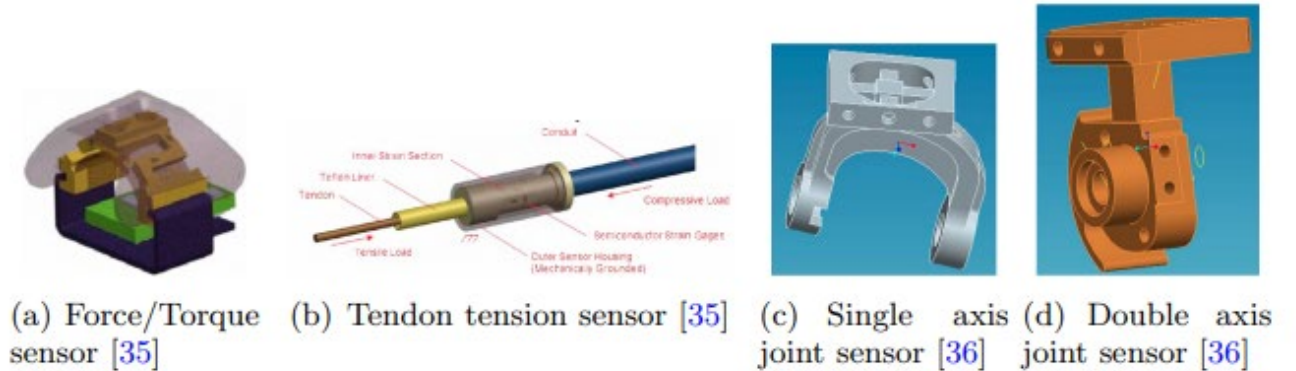
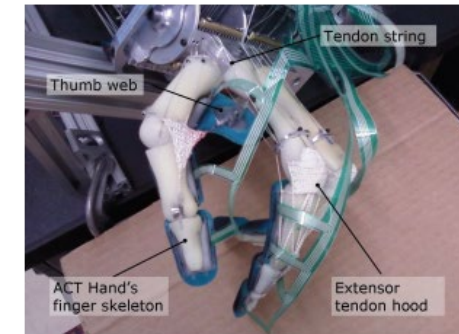


FIGURE 2.15: Force/Torque integrated sensors



(a) Sensorized artificial skin concept



(b) Integration of the skin within the robotic hand

FIGURE 2.19: Artificial skin with integrated tactile sensors

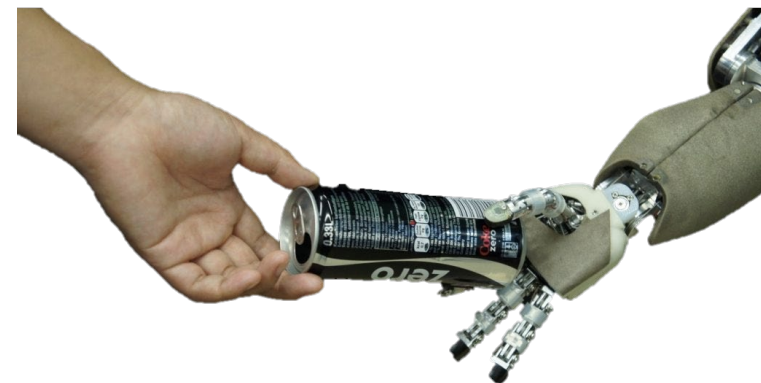
机器人力与触觉传感器 关键技术与应用

李淼

武汉大学机器人学院

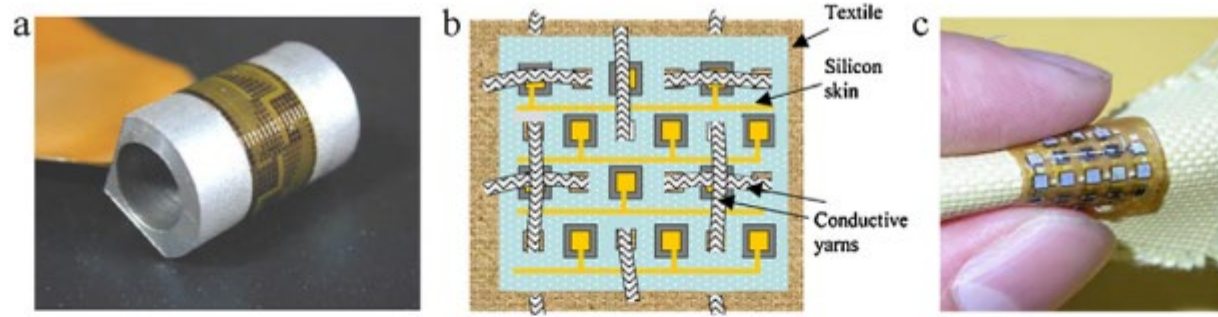
教授 博导

2026 武汉

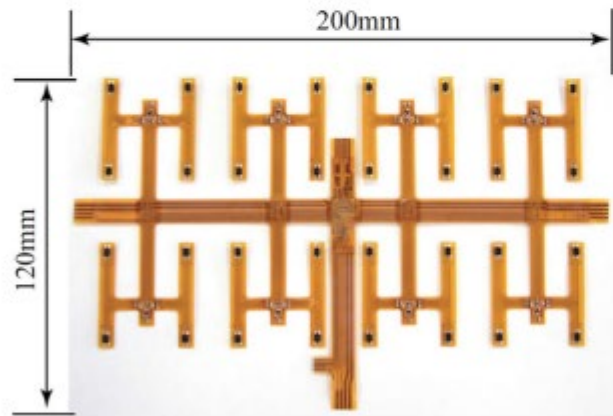




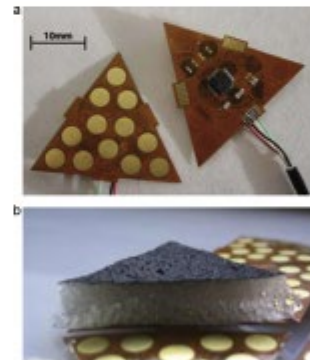
Sensors



(a) Strain gauges tactile sensor [37]



(b) Led based tactile sensor [38]



(c) Capacitive tactile sensor [39]



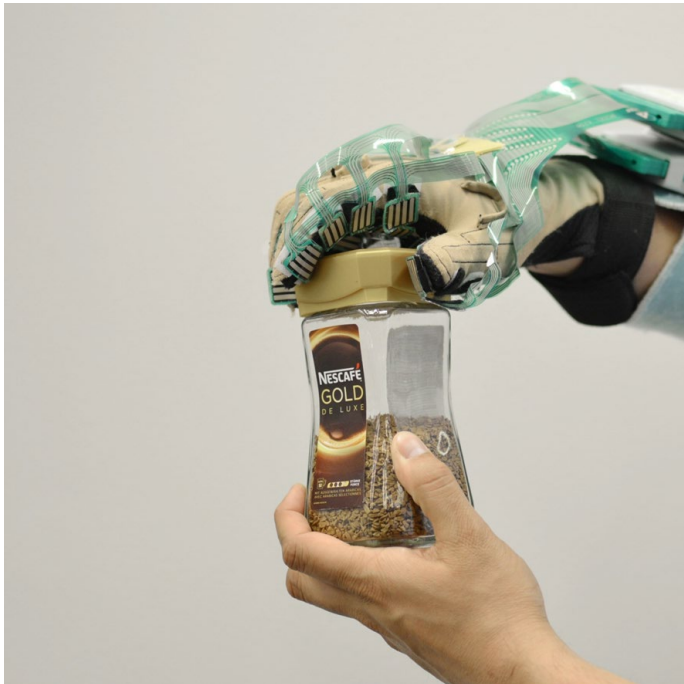
FIGURE 2.16: Tactile sensors



Sensors



FIGURE 2.3: 3D hand motion capture: CyberGlove

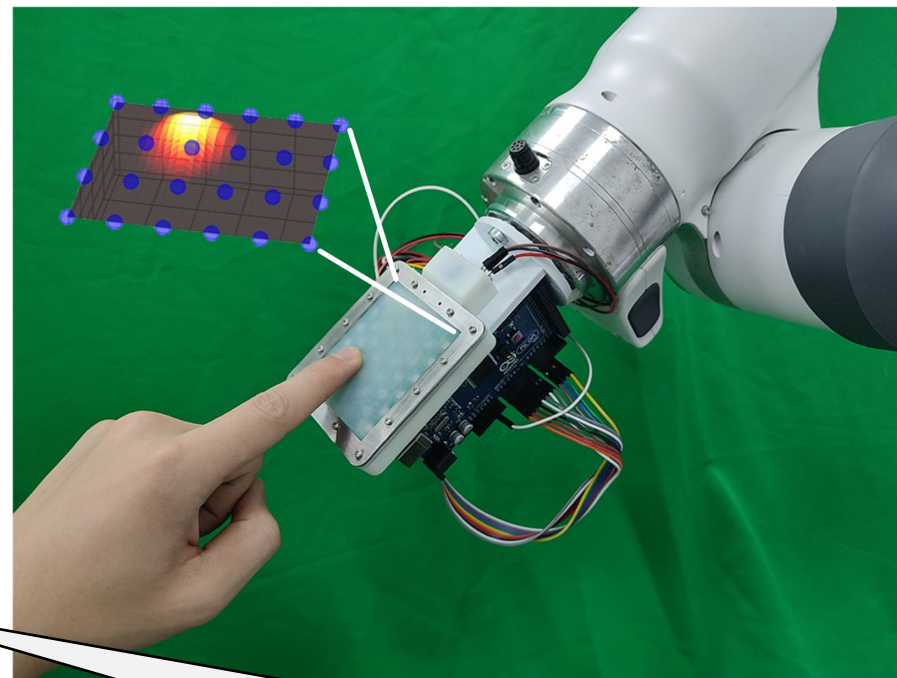


Learning Object Exploration from Human Demonstration

Dr. Sahar El Khoury
Ravin Luis De Souza
Miao Li

Prof. Aude Billard
Learning Algorithms and Systems Laboratory
EPFL

柔性触觉传感器



In case you need some tactile sensor for your design

A biomimetic tactile palm for robotic object manipulation

Ziwei Lei^{1,3}, Xutian Deng¹, Yi Wang¹, Xiaohui Xiao², Dong Han³, Fei Chen^{4,*}, Miao Li^{1,2,*}



Sensors

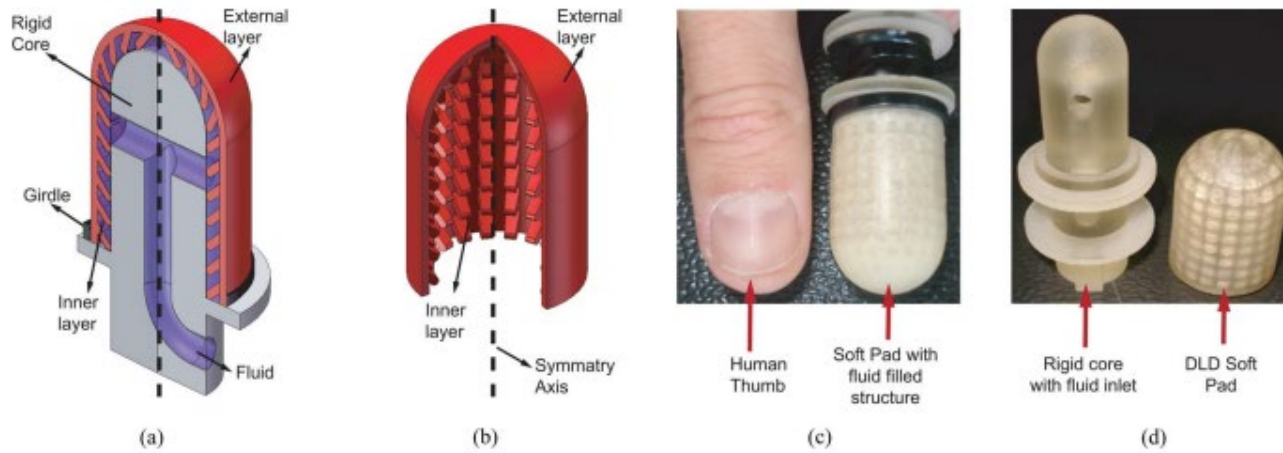


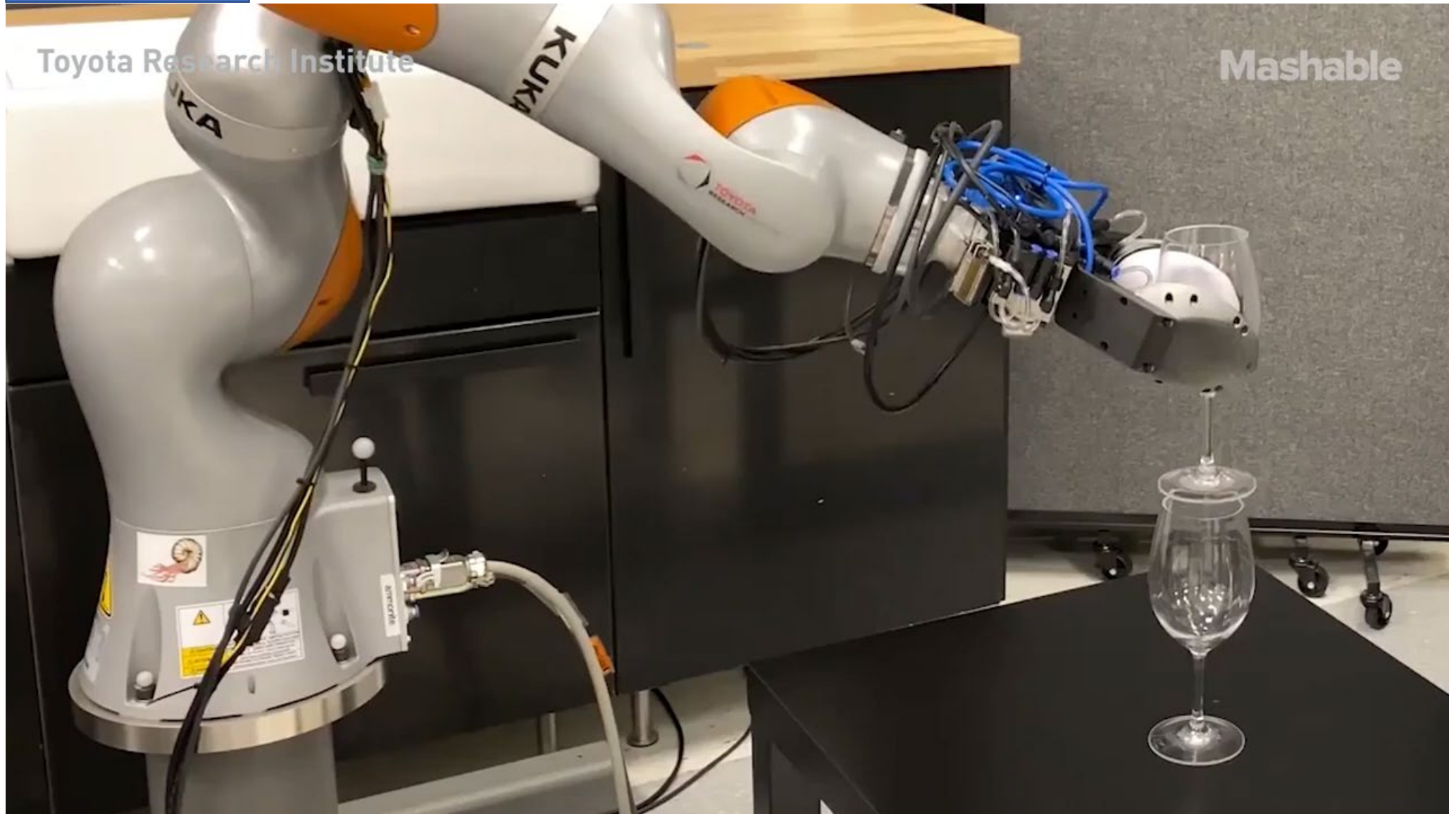
FIGURE 2.18: Fluid-filled soft-pad concept and prototype. (a) 3-D model. (b) Longitudinal cross section. (c) Prototype comparison with human-thumb dimensions. (d) Rigid core with fluid inlet and soft pad.





Video
(1 mins)

Sensors



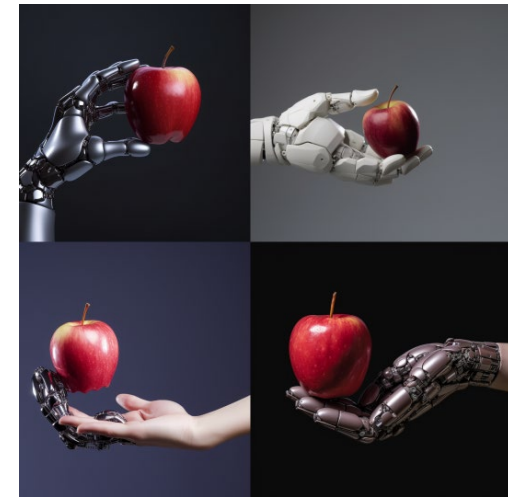
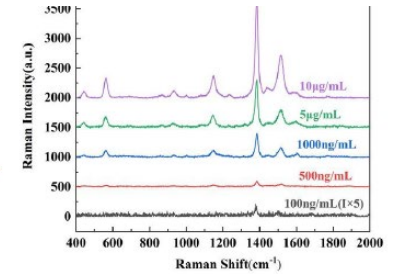
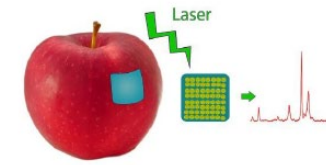
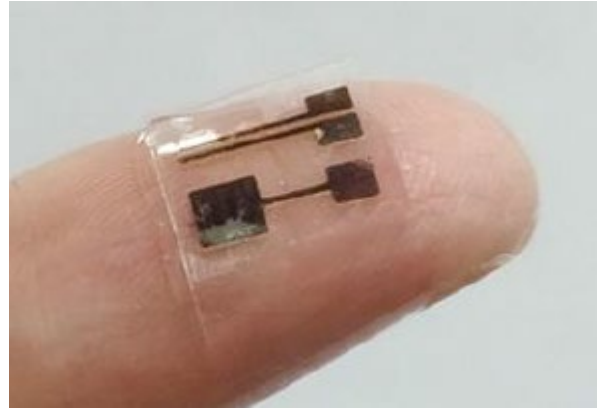


Sensors

Attach the device to the palm of your hand



Can we attach this sensor to something else?



Please join us if you are interested in this project!



Sensors

突破！灵巧手搭载嗅觉传感器，开启机器人感知新维度

手智创新 2026年1月30日 15:00 湖北

点击上方蓝字·关注我们

当灵巧手的精准操作，遇上“闻味识物”的细腻感知——近日，由手智创新团队牵头，联合湖北大学研发、华中科技大学协调整试、武汉市科创局专项支持的嗅觉检测模组，成功集成至灵巧手设备中。这一突破，让机器人首次同时拥有了“灵活的手”与“灵敏的鼻”，也为人形机器人的感知能力带来了颠覆性升级。



不止是“会握”“会抓”“会拿”，现在还“会闻”，展示嗅觉模组的黑科技





Control Algorithm

Robotic Hands

STABILITY OF VARIABLE GRASP STIFFNESS CONTROL

In this part, we address the control stability of varying stiffness in robotic grasping using a two-fingered example, as shown in Fig. B.1(a). To this end, a detailed formulation for the dynamics of the system and the soft finger model is provided. The simulation on the grasping of the object stably and from the proof of

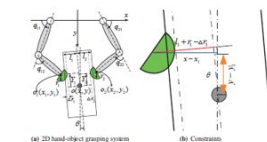


FIG. B.1: (a) The hand-object system in 2D. Each finger has 2 DOFs with soft fingertips. (b) The constraint that the fingertip should keep contact with the object's surface.

B.1 DYNAMICS

In this part, we first consider the contact model of the fingertips and the constraints involved in the hand-object system. Then the overall dynamics of the system is formulated. The notations in this section are explained in Fig. B.1 to simplify the understanding of the grasping dynamics.

B.1.1 CONTACT MODEL OF SOFT FINGERTIP

$$f = c_1(\Delta r)^2 + c_2 \frac{d}{dt} \Delta r \quad (B.1)$$

where c_1 and c_2 are positive constant parameters which depend on the material of the fingertip, and Δr is the deformation at the fingertip. The fingertip should keep contact with the object surface, as shown in Fig. B.1(b), which can be expressed as follows:

$$l_1 + r_1 - \Delta r_1 = (x - x_1) \cos \theta - (y - y_1) \sin \theta \quad (B.2)$$

$$l_2 + r_2 - \Delta r_2 = -(x - x_2) \cos \theta + (y - y_2) \sin \theta \quad (B.3)$$

B.1.2 ROLLING CONSTRAINTS

$$V_1 = (x_1 - x) \sin \theta + (y_1 - y) \cos \theta \quad (B.5)$$

$$Q_1 + Q_2 + Q_3 = \pi + \theta \quad (B.6)$$

$$Q_1 + Q_2 + Q_3 = \pi - \theta \quad (B.7)$$

B.1.3 OVERALL DYNAMICS

The total kinetic energy for the overall system can be described as follows:

$$K = \sum_{i=1}^2 \frac{1}{2} M_i \dot{q}_i^2 + \frac{1}{2} M (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} I \dot{\theta}^2 \quad (B.8)$$

where $q_i = [x_i, y_i]^T$ is the vector of finger joints and $M_i \in \mathbb{R}^{2 \times 2}$ is the inertia matrix for each finger. M and I are the mass and inertia matrix of the object respectively.

The total potential energy from deformation can be given as:

$$P = \sum_{i=1}^2 \int_0^{\Delta r_i} c_1 q_i^2 dq_i \quad (B.9)$$

Then from the Hamilton's principle, we have

$$\int_{t_0}^{t_f} [\mu(K - P) - \sum_{i=1}^2 c_2 \Delta r_i \frac{\partial \Delta r_i}{\partial \dot{X}}] \delta X + \sum_{i=1}^2 \lambda_i [(r_1 - \Delta r_1) \frac{\partial \Delta r_1}{\partial \dot{X}} + \frac{\partial f_1}{\partial \dot{X}}] \delta X + \sum_{i=1}^2 u_i^T \delta q_i] \delta t = 0 \quad (B.10)$$

where $X = [q_1^T, q_2^T, x, y, \theta]^T$.

Then we have the overall dynamics for the object-hand system as follows:

$$H_0(q_0) \ddot{q}_0 + \frac{1}{2} \ddot{B}_1 + S_1 \dot{q}_0 + f_1 \frac{\partial \Delta r_1}{\partial \dot{q}_0} - \lambda_1 [(r_1 - \Delta r_1) \frac{\partial \Delta r_1}{\partial \dot{q}_0} + \frac{\partial f_1}{\partial \dot{q}_0}] = u_0 \quad (B.11)$$

110

$$M \ddot{x} + \sum_{i=1}^2 [f_i \frac{\partial \Delta r_i}{\partial \dot{x}} - \lambda_i \frac{\partial f_i}{\partial \dot{x}}] = 0 \quad (B.12)$$

$$M \ddot{y} + \sum_{i=1}^2 [f_i \frac{\partial \Delta r_i}{\partial \dot{y}} - \lambda_i \frac{\partial f_i}{\partial \dot{y}}] = 0 \quad (B.13)$$

$$I \ddot{\theta} + \sum_{i=1}^2 [f_i \frac{\partial \Delta r_i}{\partial \dot{\theta}} - \lambda_i [(r_1 - \Delta r_1) \frac{\partial \Delta r_1}{\partial \dot{\theta}} + \frac{\partial f_i}{\partial \dot{\theta}}]] = 0 \quad (B.14)$$

With the identities in section B.5, the overall dynamics can be simplified as

$$M \ddot{x} - (f_1 - f_2) \cos \theta + (\lambda_1 + \lambda_2) \sin \theta = 0 \quad (B.15)$$

$$M \ddot{y} - (f_1 - f_2) \sin \theta + (\lambda_1 + \lambda_2) \cos \theta = 0 \quad (B.16)$$

$$I \ddot{\theta} - f_1 Y_1 + f_2 Y_2 + l_1 \lambda_1 - l_2 \lambda_2 = 0 \quad (B.17)$$

$$I \ddot{\theta} - f_1 Y_1 + f_2 Y_2 + l_1 \lambda_1 - l_2 \lambda_2 = 0 \quad (B.18)$$

B.2 VARIABLE GRASP STIFFNESS CONTROL

Motivated by the analysis of the overall system dynamics, the following control law is adopted for each finger to achieve stable grasp:

$$\ddot{q}_i = -D_i \dot{q}_i + k_d \ddot{q}_d (q_i - x_d) \quad (B.19)$$

$$x_d = \begin{bmatrix} x_d \\ y_d \end{bmatrix} \quad x_d = \frac{1}{2} \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \quad (B.20)$$

where D_i is a diagonal positive definite matrix representing the damping gain, $k \in \mathbb{R}^+$ represents the variable stiffness for each fingertip (k is the same value for the two-finger grasp to ensure force balance).

B.3 STABILITY PROOF-1

Taking the sum of inner product of Eq. (B.15) with $\dot{q}_k$, $k = 1, 2$, Eq. (B.16) with $\dot{y}$, Eq. (B.17) with $\dot{\theta}$, Eq. (B.18) with $\dot{\theta}$, we have

$$\frac{d}{dt} E = - \sum_{i=1}^2 [\dot{q}_i^T D_i \dot{q}_i + c_2 \Delta r_i^2] + \frac{1}{2} (\lambda_1 - \lambda_2)^T (x_1 - x_2) \quad (B.21)$$

$$E = K + V + P \quad (B.22)$$

111




Surprisingly Robust In-Hand Manipulation: An Empirical Study

Aditya Bhatt* Adrian Sieler* Steffen Puhlmann Oliver Brock

Presented at RSS 2021

* Both authors contributed equally to this work.

Model-based → Learning-based



Application Scenarios

Video
(1 mins)

Object Motion Primitives

Translation

Rotation

Range

X: ± 25 mm

Y: ± 25 mm

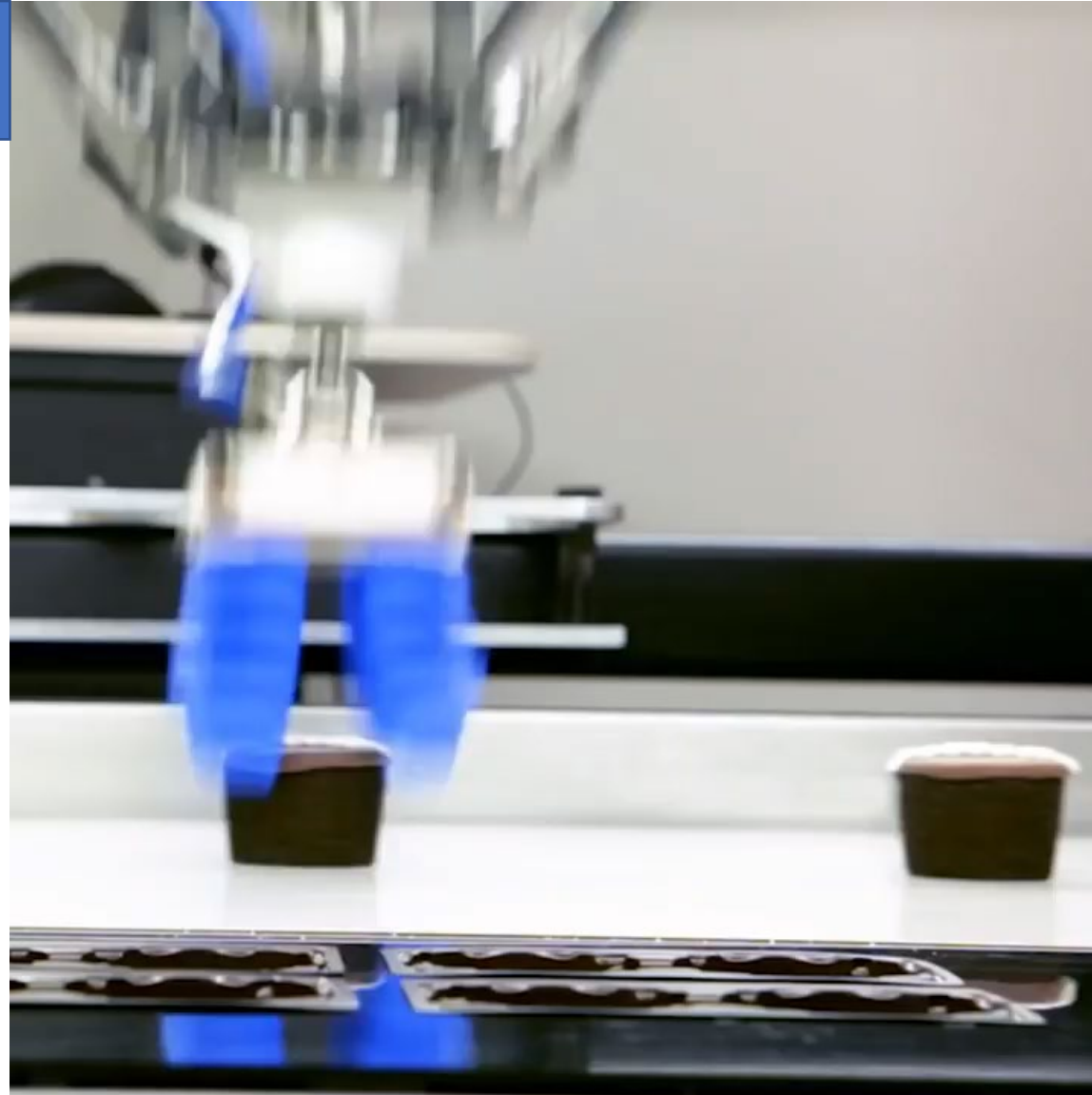
▶▶ 4x





Application Scenarios

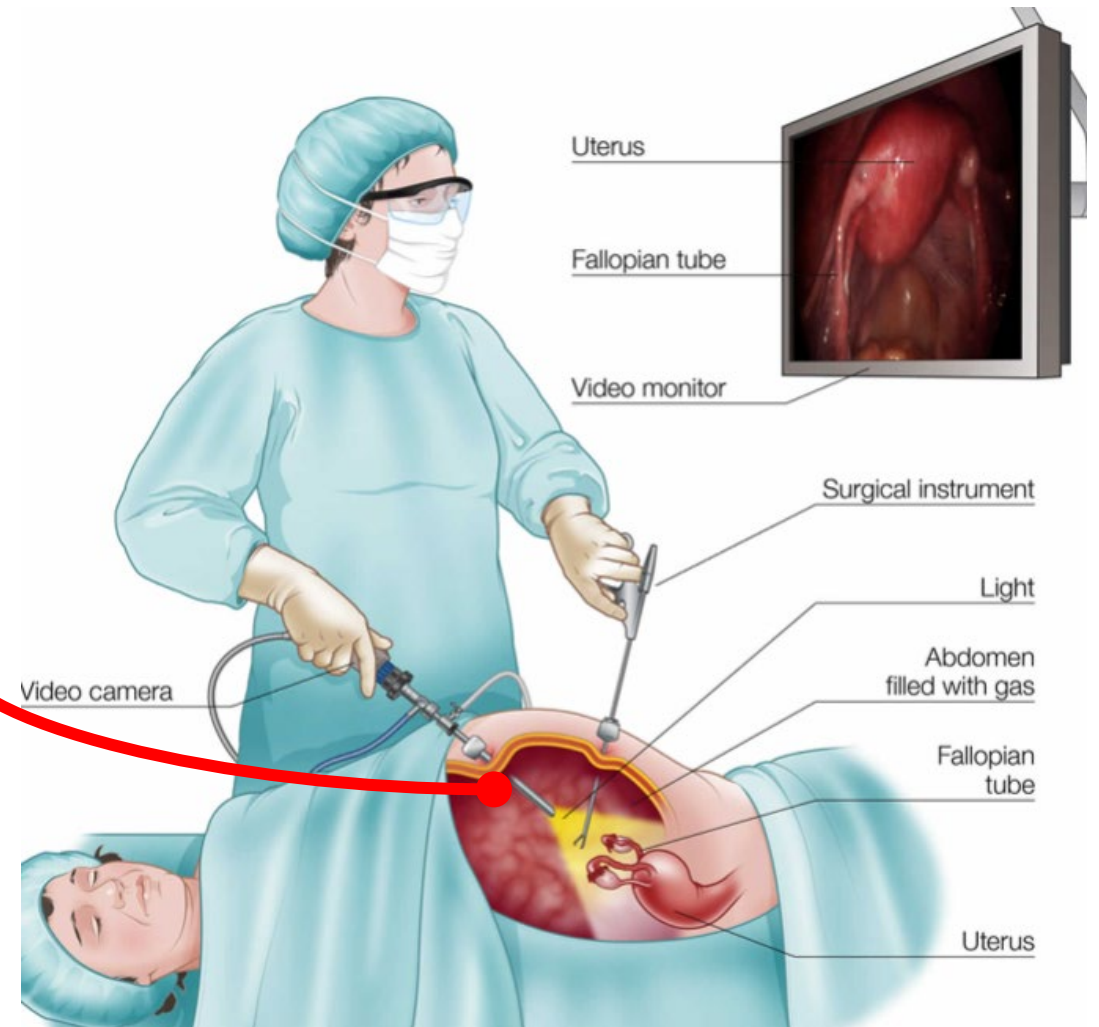
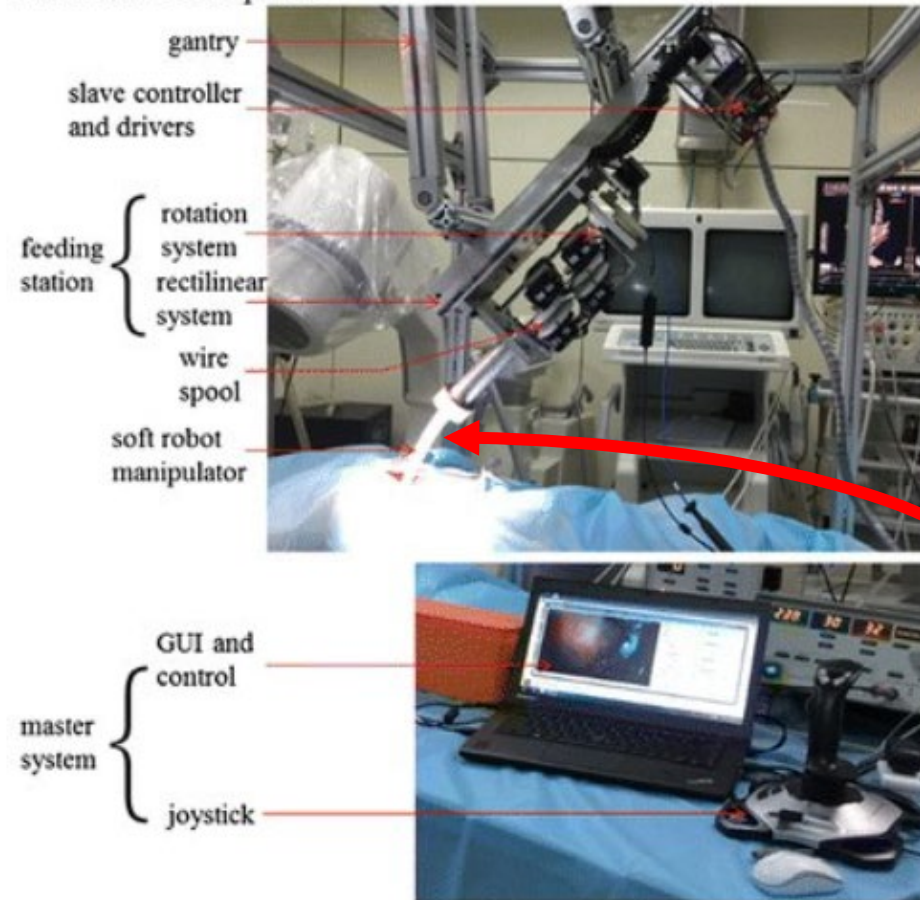
Video
(1 mins)





Application Scenarios

Hardware description





Application Scenarios





Application Scenarios



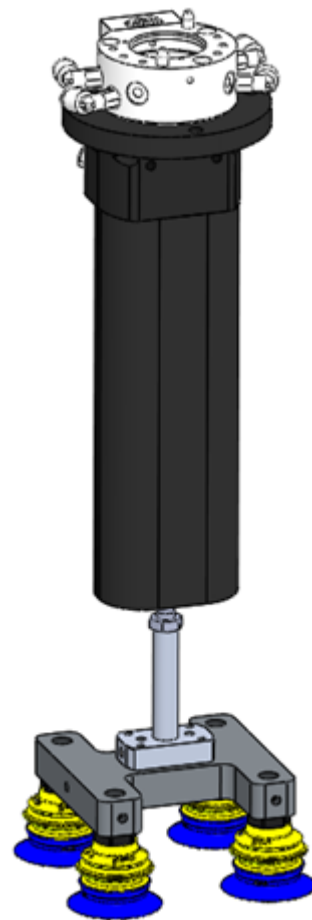
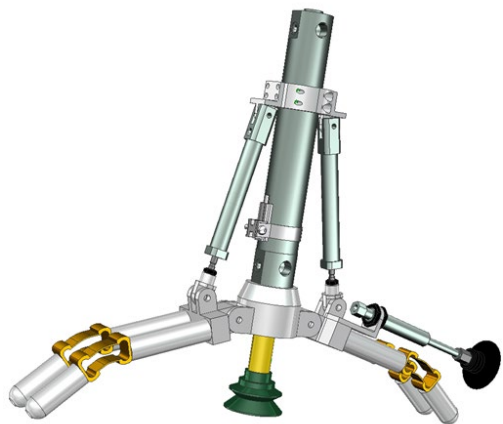
Can we design toys with soft hands?

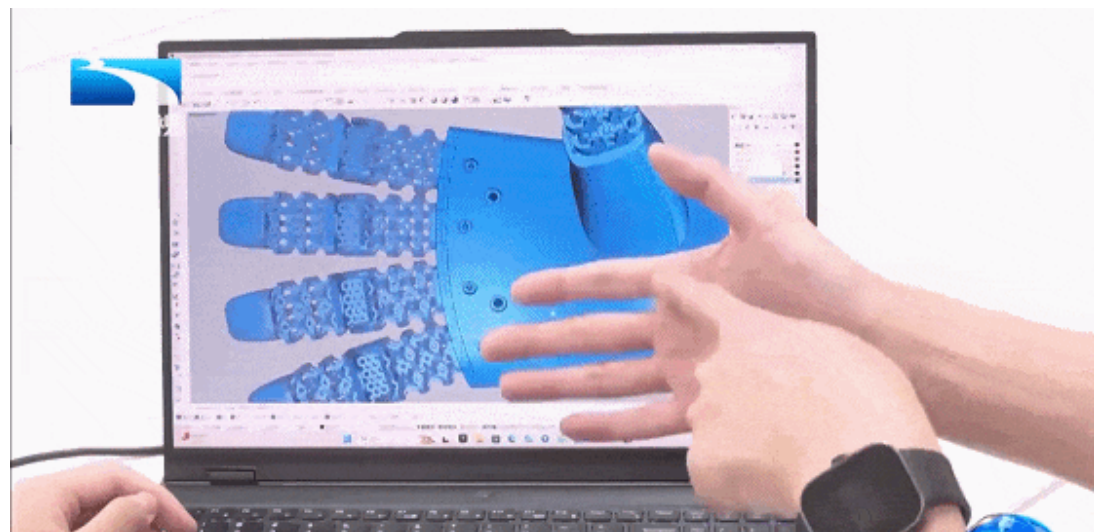
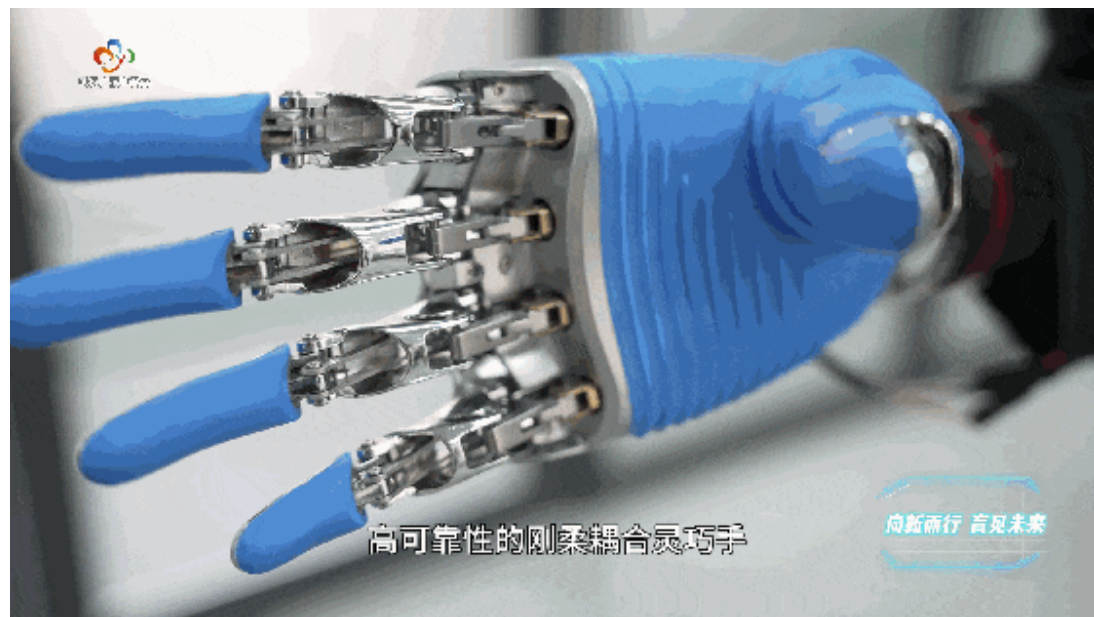


Application Scenarios



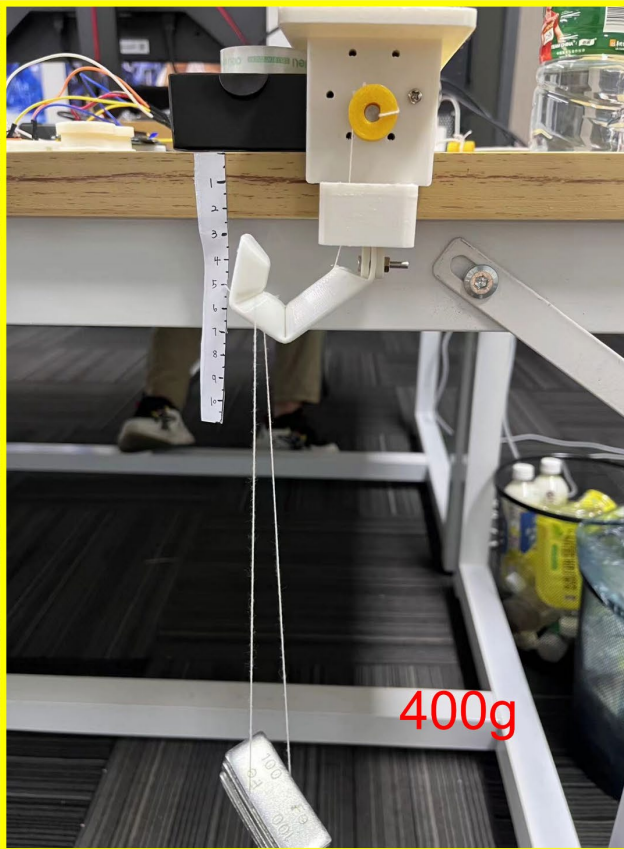
Soft hands for driving?

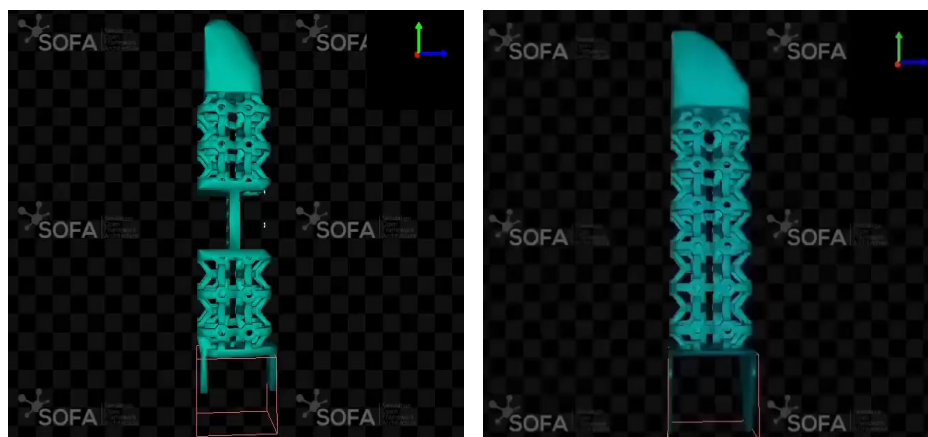
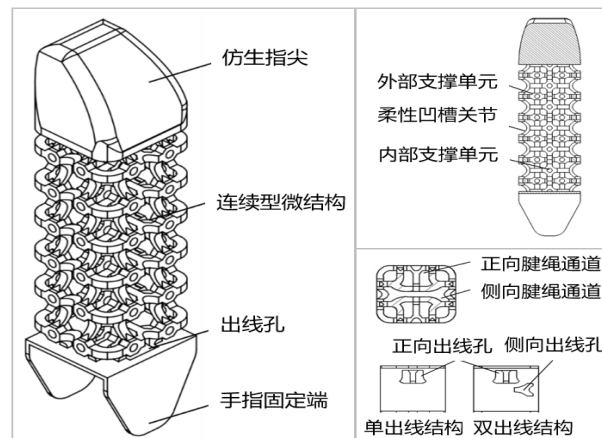
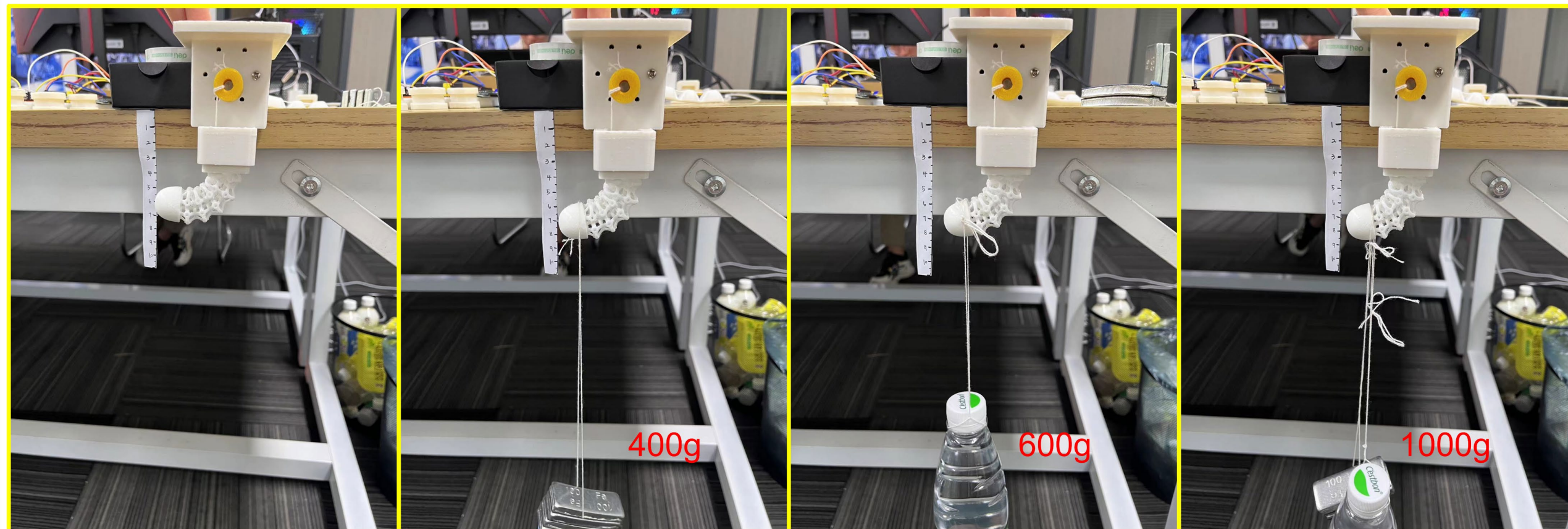


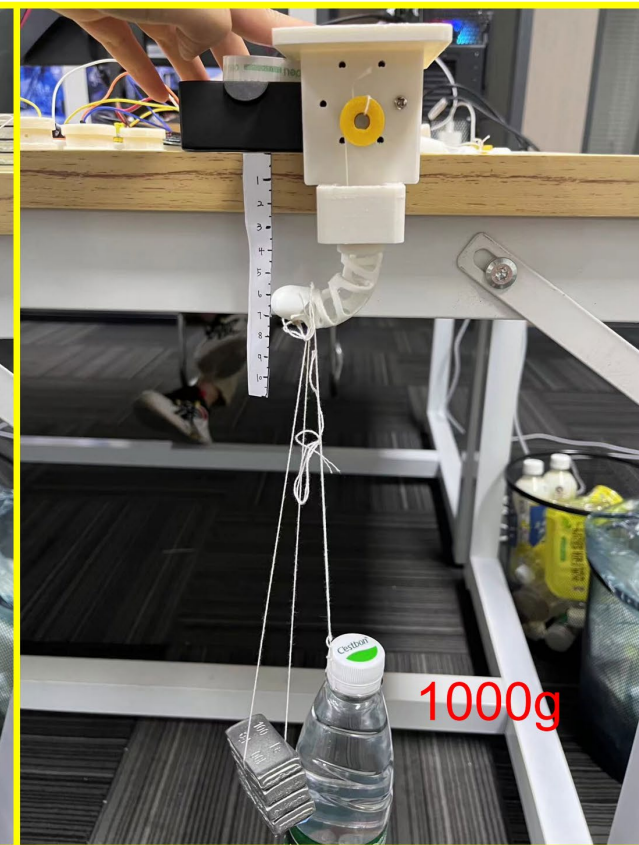
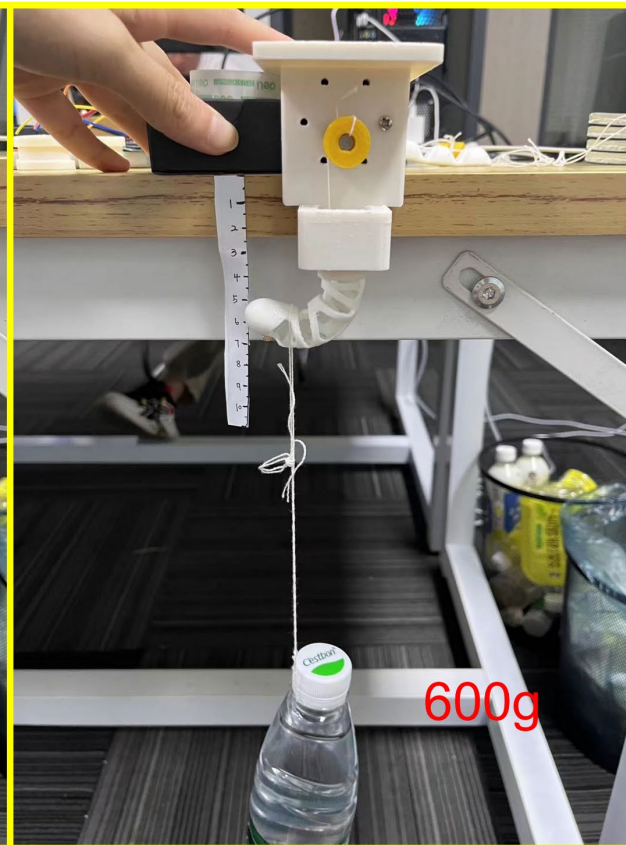
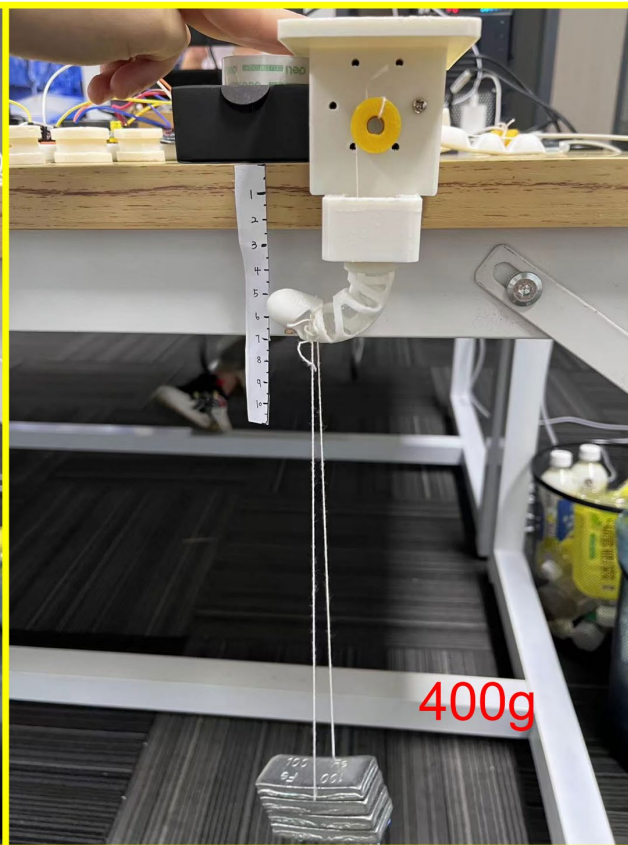


A



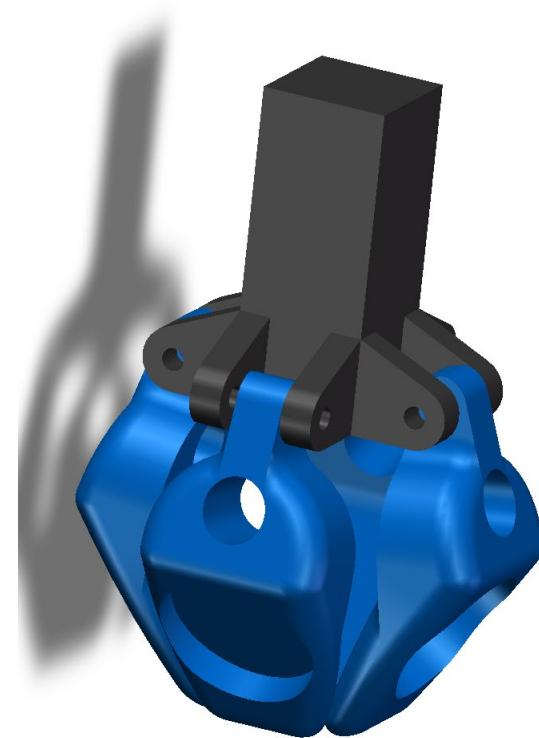
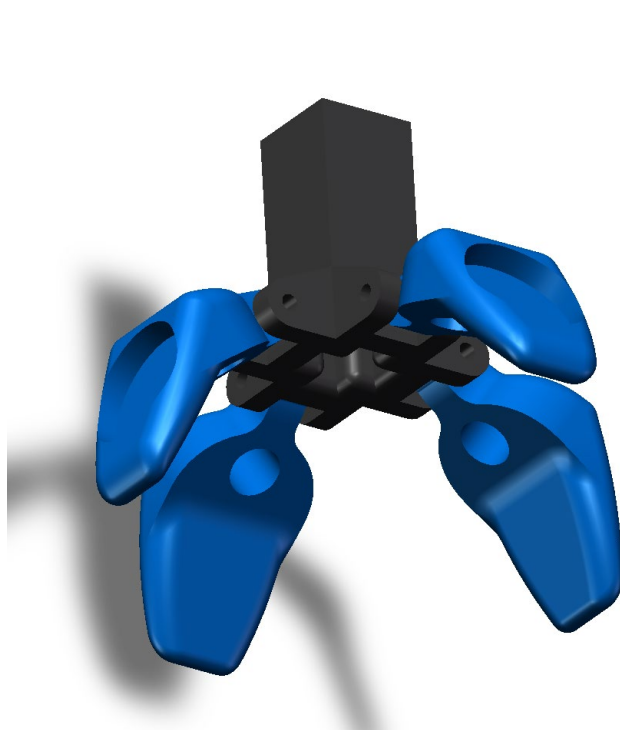
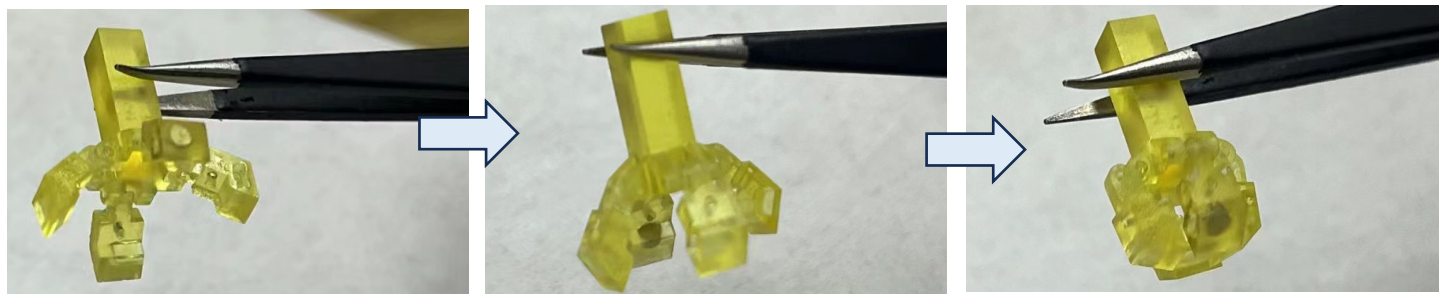
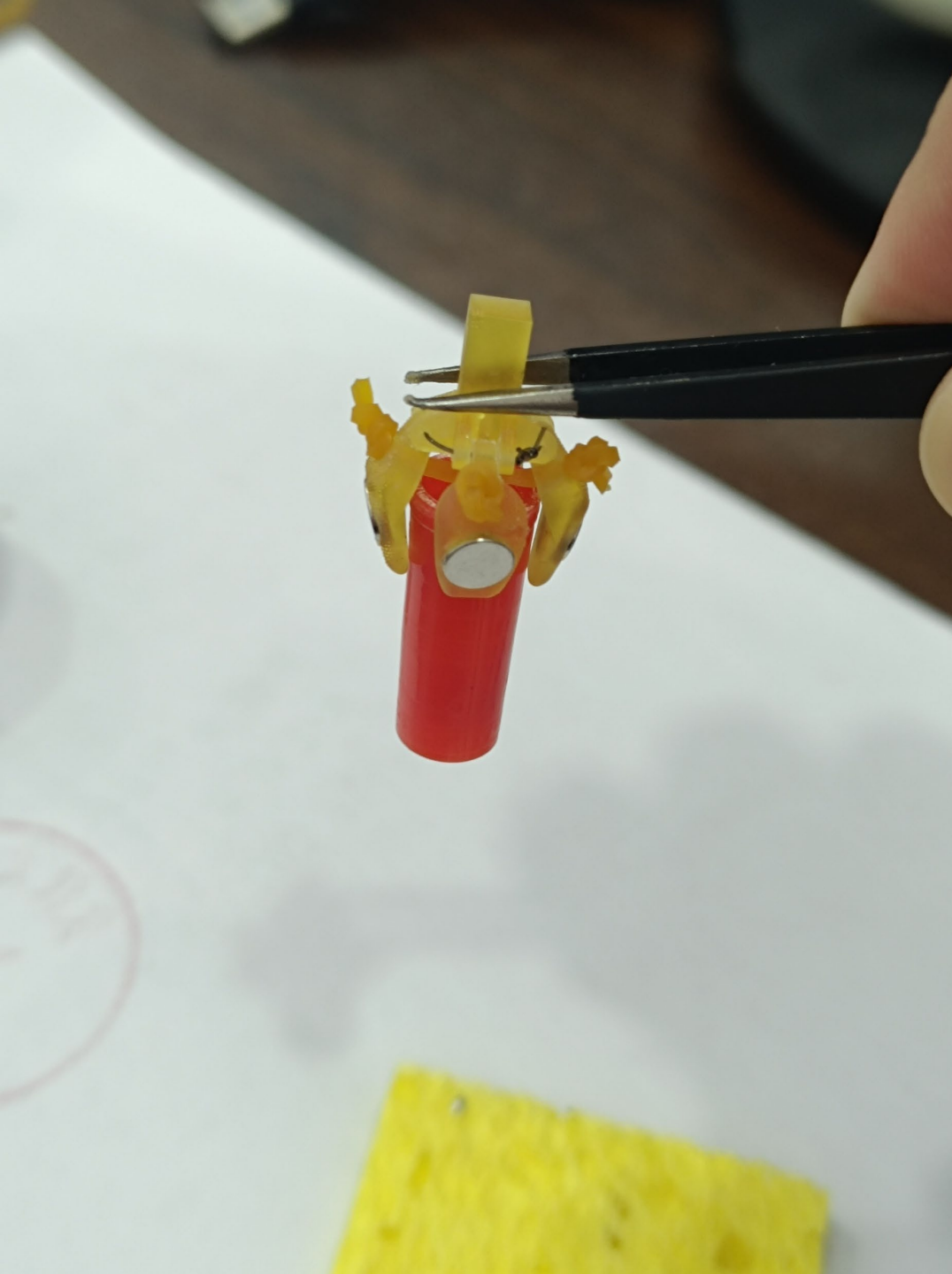






Tactile Data Glove







ICRA

May 30 to June 5, 2021

Xian China



Soft Robotics for Delicate and Dexterous Manipulation

Robert Wood
Harvard University

ICRA 2021 Plenary Talk



https://www.youtube.com/watch?v=gskX0cwCTZQ&ab_channel=IEEERoboticsandAutomationSociety

IEEE 2021
ICRA
May 30 to June 5, 2021
Xi'an China



Soft Robotics for Delicate and Dexterous Manipulation

Robert Wood
Harvard University

ICRA 2021 Plenary Talk
June 1, 2021

